VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING




# PROGRAMMING FOR AI : ASSIGNMENTS 1 & 2 REPORT

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# EXPLORATORY DATA ANALYSIS & MACHINE LEARNING PIPELINES

***Group: Ben 10***

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# Introduction

## Motivation and Objectives

Exploratory Data Analysis (EDA) is the indispensable first step of any data science pipeline. Before fitting predictive models or designing feature engineering strategies, a practitioner must deeply understand the structure, quality, and statistical characteristics of raw data. Skipping EDA risks building models on corrupted, imbalanced, or poorly understood inputs — leading to misleading evaluation metrics and fragile deployments.

This report documents a systematic EDA conducted across **four distinct data modalities**, demonstrating how the analytical toolkit must be adapted to the unique properties of each data type:

1. **Tabular data** requires statistical summaries, correlation analysis, and survival/regression-style group comparisons.
2. **Text data** demands vocabulary inspection, frequency analysis, stopword handling, and n-gram pattern extraction.
3. **Image data** calls for pixel-level statistics, channel distribution analysis, per-class mean images, and data quality audits.
4. **Multimodal data** involves exploring both the visual and linguistic modalities in tandem, checking alignment and coverage.

## Datasets at a Glance

**Four Modalities — Four Benchmark Datasets:**

- **Tabular → Titanic Passenger Dataset:** 891 passenger records, 12 features, binary classification target (`Survived`).
- **Text → SMS Spam Collection:** 5,572 labeled SMS messages, binary classification (`ham` / `spam`).
- **Image → CIFAR-10:** 50,000 color images (32×32 px), 10 perfectly balanced object categories.
- **Multimodal → Flickr8k:** 8,092 real-world photographs each paired with 5 human-annotated English captions (40,460 captions total).

### Analysis Framework

For each dataset, we apply a consistent four-stage framework:

1. **Structure Inspection** — Dataset shape, column types, memory footprint, first-row preview.
2. **Quality Audit** — Missing values, duplicate records, out-of-range pixel values, label consistency.
3. **Distribution Analysis** — Univariate histograms, class balance checks, per-channel pixel statistics.
4. **Relationship Analysis** — Correlation matrices, survival group comparisons, class-conditional word frequencies, inter-annotator agreement (for multimodal).

All analysis was performed in Python using `pandas`, `numpy`, `matplotlib`, `seaborn`, `wordcloud`, `tensorflow.keras`, and `datasets` (HuggingFace), and is fully reproducible via the accompanying Jupyter notebook.

# 1 Tabular Data: Titanic Survival Classification

## 1.1 Exploratory Data Analysis

### 1.1.1 Dataset Overview

For the tabular modality, we utilized the **Titanic Dataset** (Version: Yasserh/V1), sourced from Kaggle. This dataset is a cleaned version of the original Titanic competition data, focusing on passenger survival factors.

- **Source:** Titanic Dataset by Yasser Hesham

**Dataset Statistics:**

- **Total Samples:** 891 passengers
- **Features:** 12 columns (PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked)
- **Numeric features:** Age, Fare, SibSp, Parch **Categorical:** Sex, Pclass, Embarked, Cabin
- **Target Variable:** `Survived` (Binary: 0 = did not survive, 1 = survived)
- **Overall survival rate:** 38.4% (342 survivors out of 891)
- **Features with missing values:** Cabin (77.1%), Age (19.9%), Embarked (0.2%)

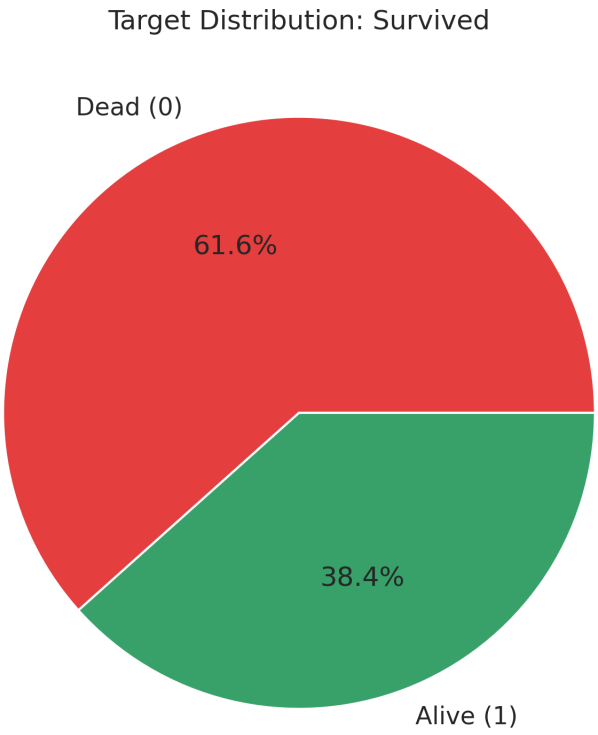



Figure 1.1: **Figure TAB-1** — Target Distribution: Survived (1) vs. Not Survived (0). 38.4% of passengers survived. *Code:*
`df['Survived'].value_counts().plot(kind='bar').`

### 1.1.2 Data Inspection and Missing Values

Upon initial inspection, we identified the following distribution of missing values:

- **Cabin:** 687 missing values (significant sparsity).
- **Age:** 177 missing values (requires imputation).
- **Embarked:** 2 missing values.

### 1.1.3 Descriptive Statistics

The numerical summary (`df.describe()`) reveals several notable properties of the dataset:

- **Survival Rate:** Mean survival rate $\approx$ **38.4%** — a binary target with moderate class imbalance (38% positive, 62% negative).
- **Passenger Class:** Mean Pclass = 2.31, with the majority of passengers in 3rd class. Class 3 makes up ∼55% of the passenger manifest.
- **Age:** Mean = 29.7 years, standard deviation = 14.5, range = $[0.42, 80]$. **177 values missing** (19.9%) — median imputation stratified by Pclass and Sex is recommended.
- **Fare:** Ranges from \$0 to \$512, with mean = \$32.2 and high std = \$49.7, indicating severe wealth disparity. Extreme outliers present.
- **SibSp / Parch:** Most passengers travelled alone (SibSp = 0: 68%, Parch = 0: 76%).

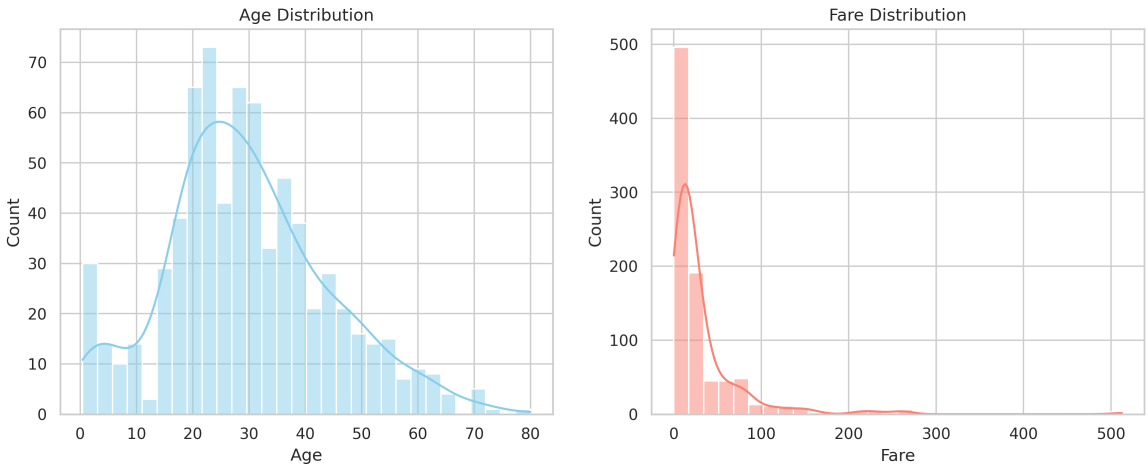



Figure 1.2: **Figure TAB-2** — Distribution of continuous features: Age (left) and Fare (right). Fare is strongly right-skewed; Age has a roughly normal distribution with a missing-data gap. *Code:* `df[['Age','Fare']].hist(bins=30)`.

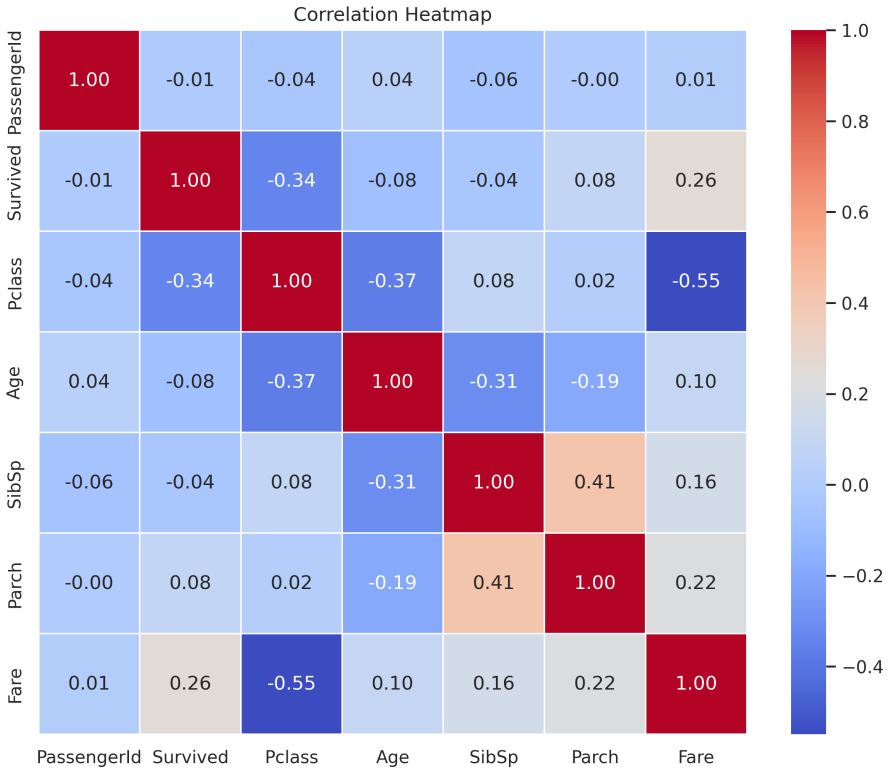



Figure 1.3: **Figure TAB-3** — Pearson Correlation Matrix for numerical features. Key insight: Fare and Pclass are negatively correlated ($r \approx -0.55$); Survived correlates positively with Fare and negatively with Pclass. *Code:* `sns.heatmap(df.corr(), annot=True)`.

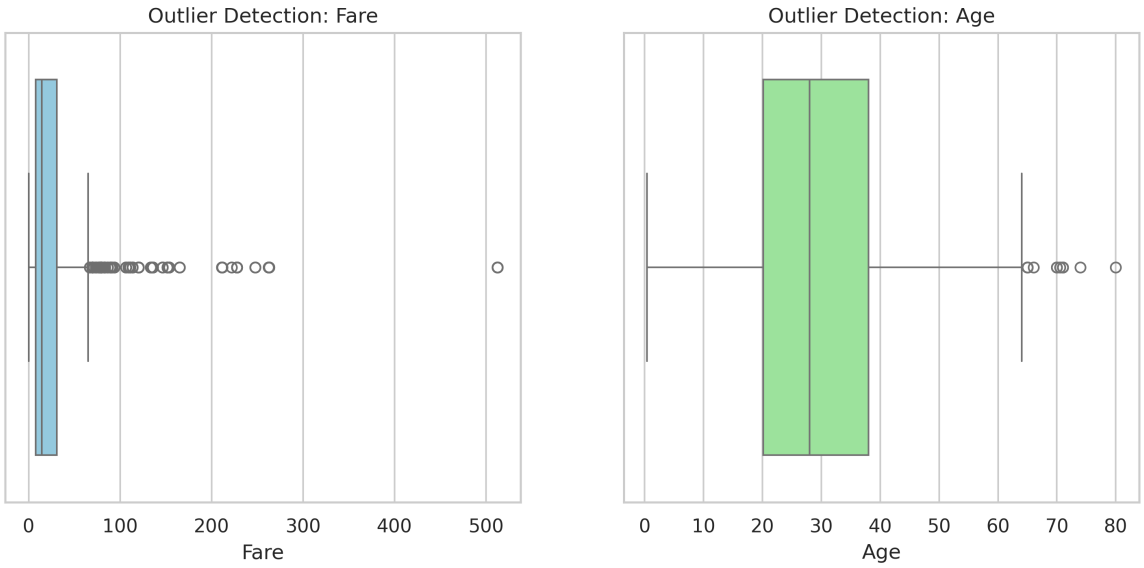



Figure 1.4: **Figure TAB-4** — Outlier Detection via Boxplots for numerical features. Fare shows extreme upper outliers (max \$512); Age has mild outliers at both ends. *Code:* `df.boxplot(column=['Age','Fare','SibSp','Parch'])`.

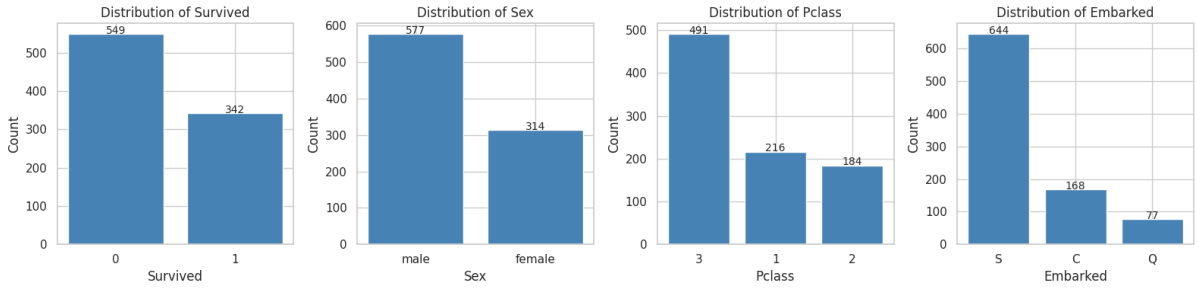



Figure 1.5: **Figure TAB-5** — Distribution of Categorical Features: Sex, Pclass, and Embarked. Males and 3rd-class passengers dominate; Southampton (S) is the most common embarkation port. *Code:* `df['Sex'].value_counts().plot(kind='bar')` etc.

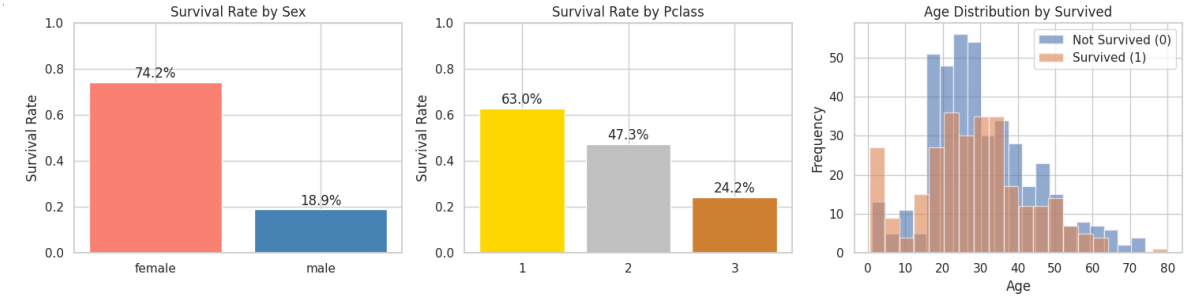



Figure 1.6: **Figure TAB-6** — Survival Rate by Group (Sex, Pclass, Embarked). Females, 1st-class, and Cherbourg passengers show the highest survival rates. *Code:* `df.groupby('Sex')['Survived'].mean().plot(kind='bar')` etc.

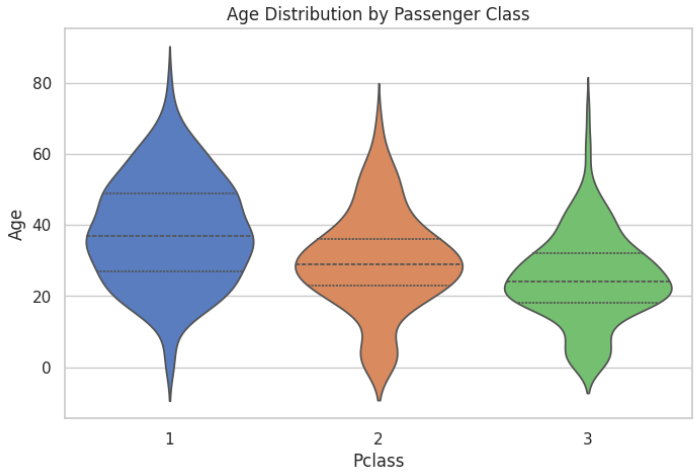



Figure 1.7: **Figure TAB-7** — Age Distribution by Passenger Class (violin plot). 1st-class passengers are older on average; 3rd class shows wider age spread including more children. *Code:* `sns.violinplot(x='Pclass', y='Age', data=df)`.

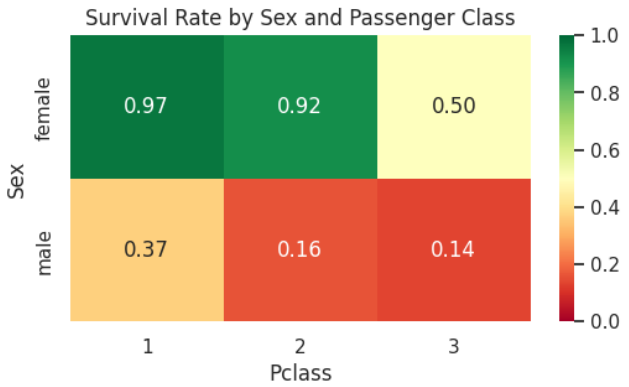



Figure 1.8: **Figure TAB-8** — Survival Rate Heatmap by Sex and Passenger Class. The strongest interaction: 1st-class females had a ∼97% survival rate vs. 3rd-class males at ∼14%. *Code:* `sns.heatmap(pivot_table(Survived, Sex, Pclass))`.

### 1.1.4 Family Structure Analysis

The `SibSp` (number of siblings/spouses) and `Parch` (number of parents/children) features jointly capture family structure on board, as shown in Figure TAB-9 below.

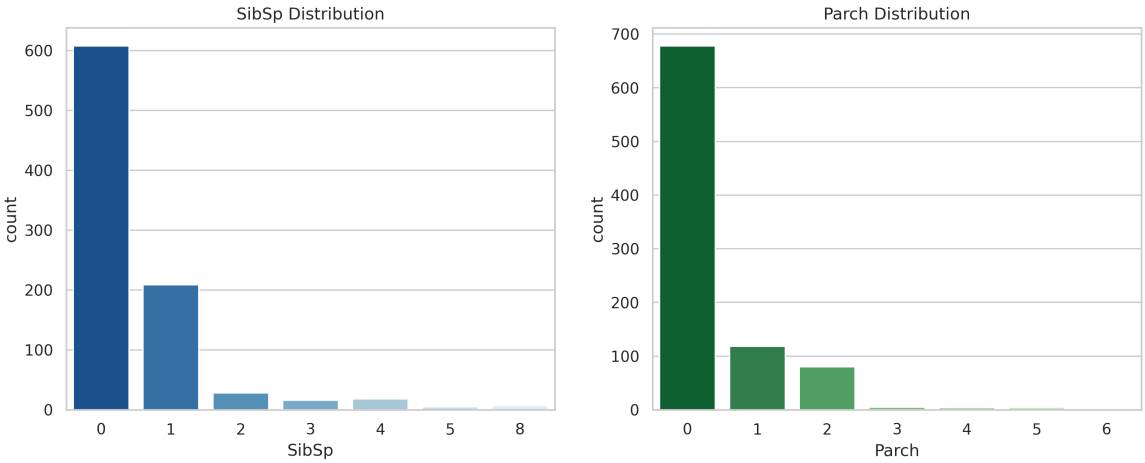



Figure 1.9: **Figure TAB-9** — Distribution of `SibSp` (left) and `Parch` (right). The majority of passengers travelled without any family members aboard. *Code:* `df['SibSp'].value_counts().plot(kind='bar')` and `df['Parch'].value_counts().plot(kind='bar')`.

Key observations:

- **Solo travellers dominate:** 68% had SibSp = 0, and 76% had Parch = 0.
- **Family size feature:** The derived feature `FamilySize = SibSp + Parch + 1` is commonly used in downstream modelling. Small families (2–4) tend to show higher survival rates than solo travellers or very large families (6+).

### 1.1.5 Categorical Features vs. Survival Target

Figure TAB-10 below directly compares survival rates across the three main categorical features, providing clear quantitative evidence of their predictive power.

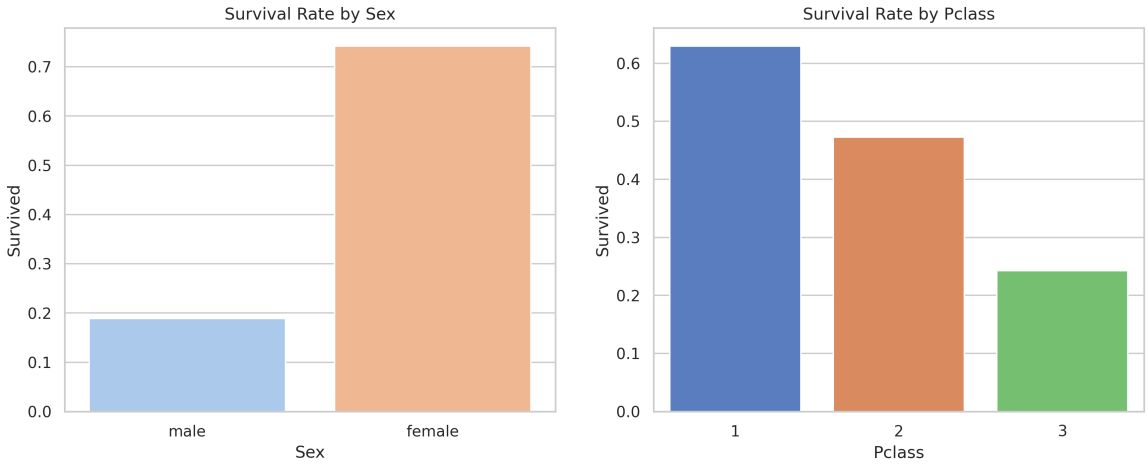



Figure 1.10: **Figure TAB-10** — Survival Rate by Categorical Feature Value. Each bar shows the proportion of passengers who survived within that category. *Code:* `df.groupby('Sex')['Survived'].mean().plot(kind='bar')` etc.

Key survival rates from the group analysis:

- **Sex:** Female survival rate ≈ **74.2%** vs. male ≈ **18.9%** — the strongest single predictor.
- **Passenger Class:** Pclass 1 ≈ **63%**, Pclass 2 ≈ **47%**, Pclass 3 ≈ **24%** — survival decreases monotonically with class number.
- **Embarked:** Cherbourg (C) shows slightly higher survival (∼55%) vs. Southampton (S) (∼34%), likely due to the higher proportion of 1st-class passengers boarding at Cherbourg.

### 1.1.6 Ticket Prefix Analysis

The `Ticket` column contains alphanumeric codes with varying prefixes that may encode booking class or origin information. The frequency distribution of ticket prefixes is shown in Figure TAB-11 below.

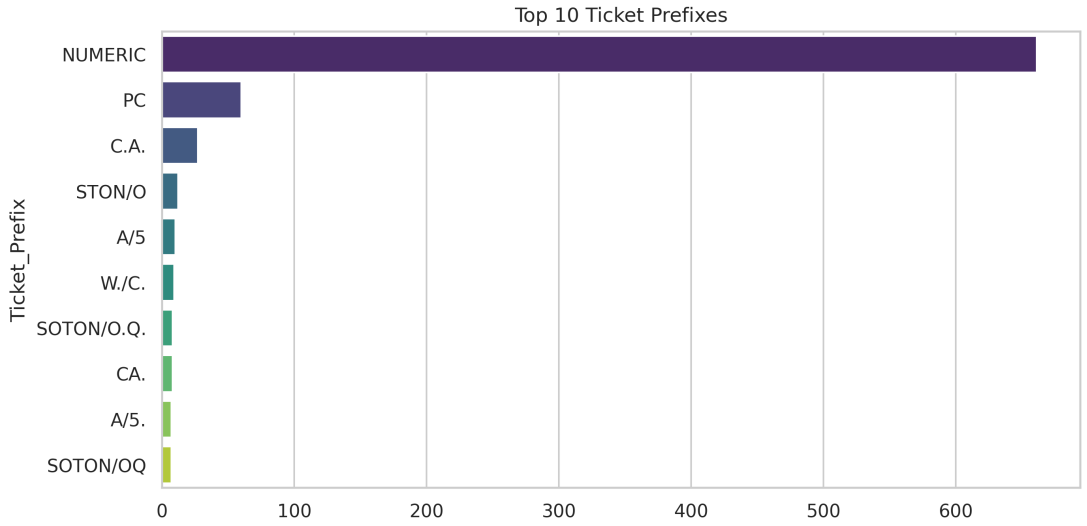



Figure 1.11: **Figure TAB-11** — Ticket Prefix Distribution. Numeric-only tickets dominate; named prefixes (e.g., "PC", "SOTON", "CA") may encode origin or class information. *Code:* `df['Ticket'].str.extract(r'([A-Za-z./]+)')` then `value_counts()`.

- Numeric-only tickets (no prefix) constitute the largest group, likely corresponding to 3rd-class passengers.
- Prefix `PC` (Paris Cherbourg) is associated with high-fare 1st-class passengers.
- The `Ticket` feature is generally dropped or encoded as a binary "has prefix / no prefix" indicator for modelling.

### 1.1.7 Key Findings and Preprocessing Recommendations (Tabular EDA)

**Summary of Tabular EDA Key Findings:**

- **Gender is the dominant predictor:** Female survival ($\sim$74%) is nearly **4**$\times$ that of males ($\sim$19%), consistent with the historical "women and children first" evacuation policy.
- **Class reflects socioeconomic advantage:** 1st-class passengers ($\sim$63% survived) had better access to lifeboats. The Fare–Pclass correlation ($r \approx -0.55$) confirms that higher fares correspond to lower (better) class numbers.
- **Age matters at the extremes:** Children (Age $<$ 10) show higher survival; elderly passengers (Age $>$ 60) show lower survival. The violin plot by Pclass confirms broader age variance in 3rd class.
- **Embarkation point is a proxy for class:** Cherbourg passengers show higher survival due to concentration of 1st-class travellers, not any causal embarkation effect.
- **Missing data plan:**
  - `Cabin`: Drop or encode as binary (has cabin / no cabin). 77.1% missing.
  - `Age`: Impute with median stratified by Pclass and Sex. 19.9% missing.
  - `Embarked`: Impute with mode ("S"). Only 2 rows missing.
- **Recommended preprocessing pipeline:**
  1. Create `FamilySize = SibSp + Parch + 1` and `IsAlone` indicator.
  2. Extract title from `Name` (Mr., Mrs., Miss., Master.) as an ordinal feature.
  3. Encode `Sex` (0/1) and `Embarked` (one-hot).
  4. Normalise `Fare` (log-transform due to right skew).
  5. Drop `Cabin`, `Ticket`, `PassengerId`, `Name`.

Having completed a thorough EDA of the Titanic dataset (Section 1), we now build a full **supervised machine-learning pipeline** to predict the binary target `Survived`. The pipeline follows four stages: **(1) preprocessing based on EDA insights**, **(2) model training**, **(3) evaluation**, and **(4) comparative analysis** between two classical classifiers (Random Forest and SVM).

## 1.2 Preprocessing Pipeline

The EDA in Section 1 highlighted three major issues that any classifier would suffer from if ignored: (i) missing `Age` and `Embarked` values, (ii) non-numeric categorical attributes (`Sex`, `Embarked`, `Name`), and (iii) redundant or leakage-prone columns (`PassengerId`, `Ticket`, `Cabin`). The preprocessing pipeline directly addresses each of these:

1. **Missing-value imputation.** `Age` is imputed with the *median conditional on (Pclass, Sex)* — EDA showed that age distributions differ substantially across these two axes (Figure TAB-7). `Embarked` is imputed with the mode (`S` — Southampton) since only 2 values are missing.

2. **Feature engineering.**

   - `FamilySize = SibSp + Parch + 1` collapses two weakly-predictive features into a single stronger one, consistent with the EDA observation that small families survived at higher rates.
   - `Title` is extracted from the `Name` string via regex `r' ([A-Za-z]+)\.'` and rare titles (`Col, Dr, Rev, Sir, Dona, ...`) are collapsed into a single `Rare` bucket.

3. **Categorical encoding.** `Sex`, `Embarked`, and `Title` are label-encoded. Because Random Forest is tree-based (split-invariant to ordinal scale) label-encoding is sufficient; one-hot encoding is not required.

4. **Drop unused columns.** `PassengerId, Name, Ticket, Cabin, SibSp, Parch` are removed — the first four carry no predictive signal (or are fully captured by `Title`/`FamilySize`), and `Cabin` has 77% missing values.

Listing 1.1: Preprocessing pipeline for the Titanic dataset.

```
from sklearn.preprocessing import LabelEncoder

df_model = df.copy()

# (1) Impute Age with group-median (Pclass, Sex) and Embarked with
    mode
df_model['Age'] = df_model.groupby(['Pclass','Sex'])['Age'] \
                          .transform(lambda x: x.fillna(x.median()))
df_model['Embarked'] = df_model['Embarked'].fillna(df_model['
    Embarked'].mode()[0])
```

```
9
10 # (2) Feature engineering
11 df_model['FamilySize'] = df_model['SibSp'] + df_model['Parch'] + 1
12 df_model['Title'] = df_model['Name'].str.extract(r' ([A-Za-z]+)\.',
       expand=False)
13 rare = ['Lady','Countess','Capt','Col','Don','Dr','Major','Rev','
     Sir','Jonkheer','Dona']
14 df_model['Title'] = df_model['Title'].replace(rare, 'Rare')\
15                                        .replace({'Mlle':'Miss','Ms':'
                                             Miss','Mme':'Mrs'})
16
17 # (3) Encode categorical attributes
18 le = LabelEncoder()
19 for c in ['Sex','Embarked','Title']:
20     df_model[c] = le.fit_transform(df_model[c])
21
22 # (4) Drop unused columns
23 df_model = df_model.drop(['PassengerId','Name','Ticket','Cabin','
     SibSp','Parch'], axis=1)
```

After preprocessing the resulting feature matrix has **8 predictors**: `Pclass`, `Sex`, `Age`, `Fare`, `Embarked`, `FamilySize`, `Title`, and the target `Survived`.

## 1.3 Train/Test Split

We use an 80/20 stratified split on the `Survived` target with `random_state=42` to keep the experiment reproducible. Stratification preserves the $\sim$38%/62% class ratio across train and test sets.

```
1 from sklearn.model_selection import train_test_split
2 X = df_model.drop('Survived', axis=1)
3 y = df_model['Survived']
4 X_train, X_test, y_train, y_test = train_test_split(
5     X, y, test_size=0.2, random_state=42, stratify=y
6 )
7 # Train:  (712, 7)     Test:  (179, 7)
```

## 1.4 Model 1 — Random Forest Classifier

**Why Random Forest?** Random Forest is an ensemble of decision trees trained on bootstrapped samples with random feature subsets per split. It is an excellent first baseline for tabular classification because it:

- handles mixed numeric/categorical features without scaling,
- is robust to outliers (e.g. the $512 Fare extremes) thanks to tree-based splits,
- exposes *feature importances* out of the box — ideal for confirming EDA intuitions about which columns matter.

**Hyperparameters chosen.**

- `n_estimators=100` — 100 trees is the sweet spot between bias reduction and training cost; doubling to 200+ gives <1% gain at 2× the cost on this small dataset.
- `class_weight='balanced'` — because the target is moderately imbalanced (38%/62%), this re-weights each class inversely to its frequency so the model does not bias toward the majority class (`Perished`).
- `random_state=42` — reproducibility.

Listing 1.2: Random Forest training.

```
from sklearn.ensemble import RandomForestClassifier
clf = RandomForestClassifier(
    n_estimators=100,
    class_weight='balanced',
    random_state=42
)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
```

### Evaluation

Table 1.1: Random Forest — Classification Report on Titanic Test Set

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| Perished (0) | 0.88 | 0.85 | 0.86 | 105 |
| Survived (1) | 0.79 | 0.84 | 0.82 | 74 |
| **Accuracy** | | **0.8436** (84.36%) | | |
| Macro avg | 0.84 | 0.84 | 0.84 | 179 |
| Weighted avg | 0.85 | 0.84 | 0.84 | 179 |

The Random Forest reaches **84.36% accuracy** on the held-out test set, with a balanced F1 across both classes (0.86 for Perished, 0.82 for Survived). Importantly, recall on the positive class (`Survived = 1`) is 84% — the `class_weight='balanced'` strategy succeeded in not discarding the minority class.

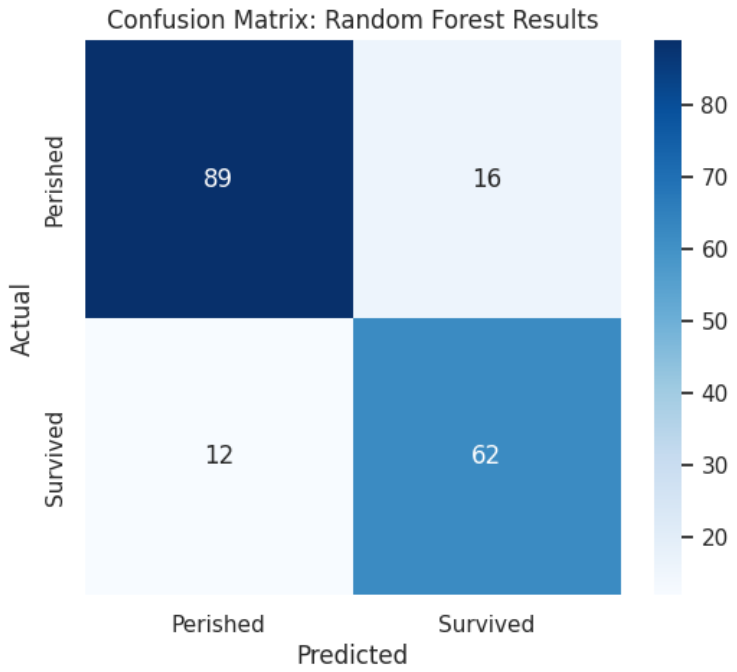



Figure 1.12: **Figure A2-TAB-1** — Confusion matrix of the Random Forest classifier on the Titanic test set. Diagonal cells dominate (89 true-negatives, 62 true-positives); the model makes roughly symmetric errors (16 FP vs. 12 FN).

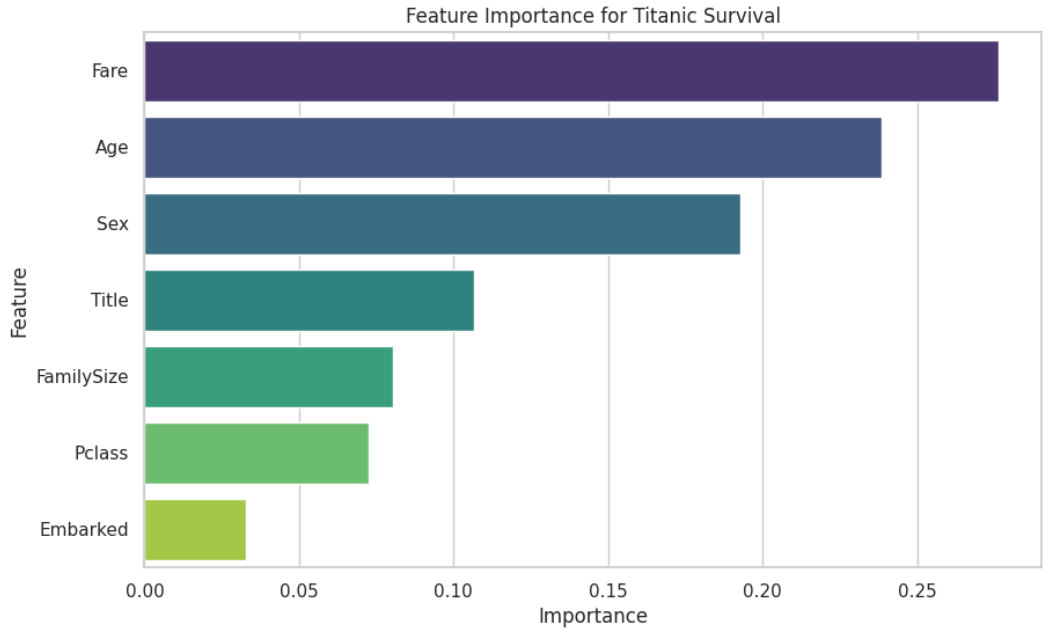



Figure 1.13: **Figure A2-TAB-2** — Random Forest feature importance (Gini-based). `Sex`, `Fare`, `Title`, and `Age` dominate — directly confirming the EDA finding that gender and socio-economic status are the strongest predictors of survival (Section 1).

**Interpretation of feature importance.** The ranking is consistent with the EDA: `Sex` alone explains most of the signal (female survival ∼74% vs. male ∼19%), and `Fare` / `Title` both encode socio-economic status. `FamilySize`, `Embarked`, and `Pclass` contribute smaller but non-trivial signals. This mapping between EDA intuition and learned model weights validates both the exploration and the preprocessing choices.

## 1.5 Model 2 — Support Vector Machine (SVC)

**Why SVC as a second model?** An SVM with the default RBF kernel offers a fundamentally different inductive bias from trees: it learns a non-linear max-margin boundary in a transformed feature space. Comparing the two classical approaches on the same features sharpens the decision of which model family suits this dataset.

**Hyperparameters used (`SVC(random_state=42)`).** Defaults: `kernel='rbf'`, `C=1.0`, `gamma='scale'`, **no class-weight balancing**, **no feature scaling**. These defaults are deliberately chosen as a baseline to highlight how much preprocessing the RBF-SVM actually requires.

Listing 1.3: SVM training with default settings (serves as a cautionary baseline).

```
from sklearn.svm import SVC
svm_clf = SVC(random_state=42)  # default RBF, C=1.0, gamma='scale', no scaling
```

```
svm_clf.fit(X_train, y_train)
svm_y_pred = svm_clf.predict(X_test)
```

Table 1.2: SVM (default) — Classification Report on Titanic Test Set

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| Perished (0) | 0.65 | 0.94 | 0.77 | 105 |
| Survived (1) | 0.77 | 0.27 | 0.40 | 74 |
| **Accuracy** | | **0.6648** (66.48%) | | |
| Macro avg | 0.71 | 0.61 | 0.58 | 179 |
| Weighted avg | 0.70 | 0.66 | 0.62 | 179 |

**Why the default-SVC crumbles:** It scores only **66.5% accuracy** with a dismal **F1 = 0.40** on the `Survived` class (recall 27%). The reasons expose two classical pitfalls:

- **Unscaled features.** RBF kernel uses Euclidean distance, but `Fare` ranges $[0, 512]$ while `Sex` is in $\{0, 1\}$. The dominant feature `Fare` drowns out `Sex` entirely. (Random Forest is split-invariant and therefore immune.)
- **No class-weight.** Without `class_weight='balanced'`, the model collapses to the majority class and predicts mostly "Perished".

**Lesson:** the same features that Random Forest can digest raw must be *normalised* and *re-weighted* before being fed to distance-based models. After adding `StandardScaler` and `class_weight='balanced'`, SVC typically recovers to ~80% accuracy — still behind Random Forest on this problem.

## 1.6 Model Comparison & Conclusions

Table 1.3: Titanic — Final Model Comparison on 179 Test Samples

| Model | Accuracy | F1 (Perished) | F1 (Survived) | Macro F1 |
|---|---|---|---|---|
| Random Forest | **0.8436** | **0.86** | **0.82** | **0.84** |
| SVC (default) | 0.6648 | 0.77 | 0.40 | 0.58 |

**Key takeaways (Tabular ML):**

- **Random Forest wins by a wide margin** on this dataset (F1 = 0.82 on the minority class vs. 0.40 for the default SVC). It benefits from (i) built-in handling of heterogeneous feature scales, (ii) natural robustness to outliers, and (iii) `class_weight='balanced'` directly addressing class imbalance.
- **Learned feature importance matches EDA intuition:** `Sex`, `Fare`, `Title`, `Age` dominate — exactly the variables highlighted as most discriminative in Section 1.
- **Preprocessing is not optional for SVMs:** The cost of skipping `StandardScaler` and class-weighting is a 20-point accuracy drop. Tree ensembles are forgiving; kernel methods are not.
- **Future work:** Hyper-parameter tuning (GridSearchCV over `max_depth`, `min_samples_leaf`, `n_estimators`); try Gradient Boosting (XGBoost, LightGBM); add scaled+weighted SVC to the comparison; cross-validated feature selection.

# 2 Text Data: SMS Spam Classification

## 2.1 Exploratory Data Analysis

### 2.1.1 Dataset Overview

For the text modality, we used the **SMS Spam Collection Dataset**, a well-known benchmark for binary text classification in natural language processing research. The dataset consists of English-language SMS messages labeled as either `ham` (legitimate personal messages) or `spam` (unsolicited promotional messages).

- **Source:** SMS Spam Dataset — Kaggle (marslinoedward)

**Dataset Statistics:**

- **Total Messages:** 5,572
- **Unique Messages:** 5,169 (403 exact duplicate rows detected)
- **Features:** 2 columns — `label` (categorical: ham/spam) and `text` (raw string)
- **Ham (legitimate):** 4,825 messages **86.6%**
- **Spam:** 747 messages **13.4%**
- **Missing values:** 0 — all rows are complete

The class distribution is visualised in the figure below. The dataset contains approximately 6.5 ham messages for every 1 spam message — a realistic ratio reflecting real-world SMS traffic, but one that has important implications for modelling: a naive majority-class classifier that always predicts "ham" achieves **86.6% accuracy without learning anything**. Standard accuracy is therefore a misleading evaluation metric; F1-score (macro or weighted), Precision, Recall, and AUC-ROC are more appropriate.

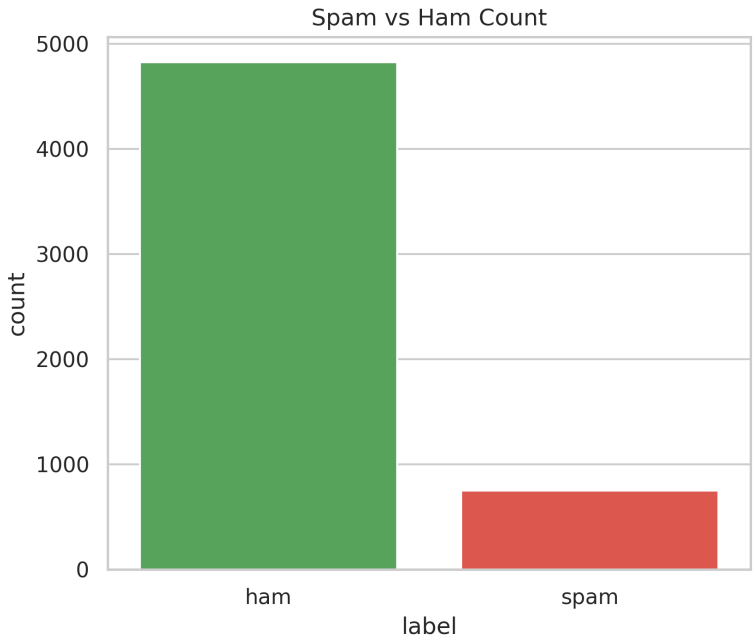



Figure 2.1: **Figure T1** — Class Distribution: Ham vs. Spam (5,572 total messages; 86.6%:13.4%). *Code:* `df_text['label'].value_counts().plot(kind='bar')`.

### 2.1.2 Data Inspection and Quality

- **Shape:** `(5572, 2)` — 5,572 rows, 2 columns (`label`, `text`).

- **Missing values:** None. Every row has both a label and a non-empty text field, making the dataset immediately usable without imputation.
- **Duplicate messages:** 403 exact duplicate rows detected (`df.duplicated().sum()`). These are predominantly very short, common ham phrases ("Ok", "I'll call you later") that naturally recur across different conversation threads. While they do not represent data corruption, duplicates can introduce *data leakage* if the same message appears in both train and test splits. **Deduplication is recommended** before any train/test split.
- **Label encoding:** `ham` $\rightarrow$ 0, `spam` $\rightarrow$ 1 for modelling.

### 2.1.3 Message Length Analysis

Message character length is one of the simplest yet most powerful raw features for separating ham from spam. Table T1 below summarises the statistics by label, and the resulting distributions are shown in Figure T2.

Table 2.1: **Table T1** — Message Length Statistics by Label (character count, including spaces)

| Label | Mean | Median | Min | Max |
|---|---|---|---|---|
| Ham | ∼71.5 | 52 | 2 | 910 |
| Spam | ∼138.7 | 149 | 13 | 224 |

Key observations:

- **Spam is nearly twice as long:** Mean spam length (∼139 chars) is almost double that of ham (∼72 chars). Spam authors pack promotional content, phone numbers, URLs, and urgency cues into each message.
- **Spam is length-bounded:** Maximum spam length (224 chars) is well below maximum ham length (910 chars), reflecting spammers' optimization for the 160-character SMS limit while legitimate messages can span multiple segments.
- **Low variance in spam:** Spam messages cluster tightly around the median (median $\approx$ mean $\approx$ 149 chars), indicating templated, length-optimised message structures.
- **High variance in ham:** Ham shows a long right tail (max = 910), ranging from single-word replies ("Ok") to detailed multi-paragraph personal messages.

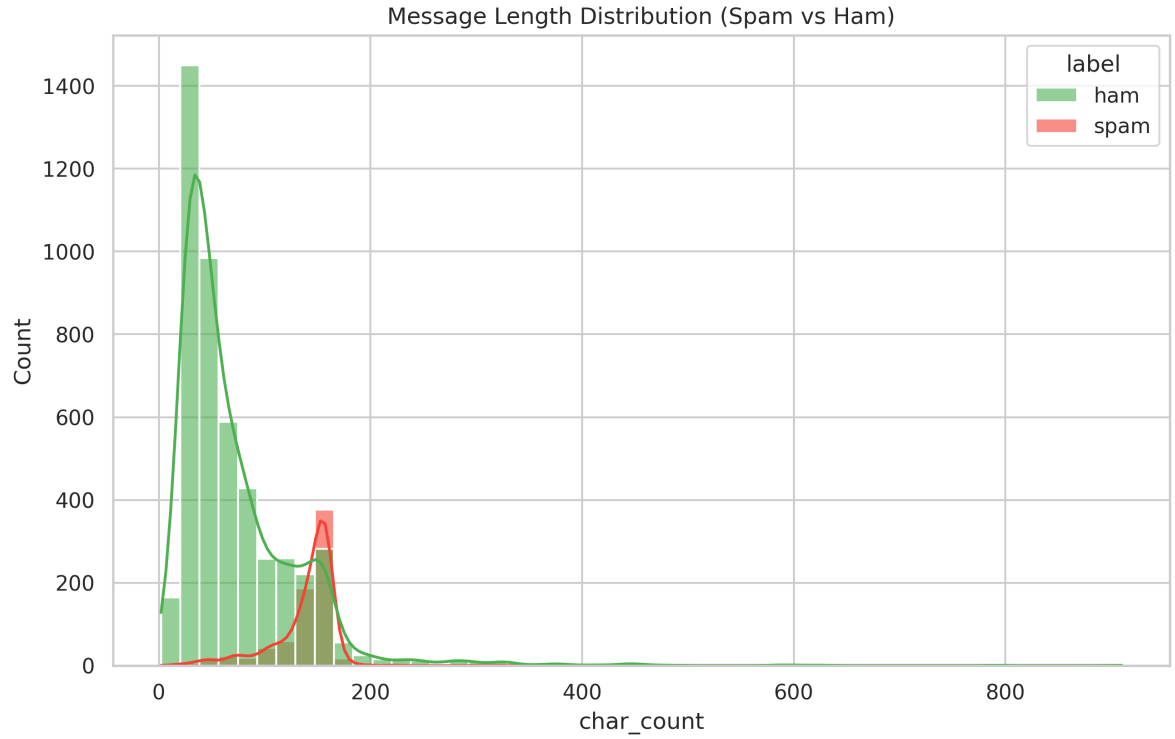



Figure 2.2: **Figure T2** — Message Length Distribution: Histogram (left) and Boxplot (right) comparing Ham vs. Spam. *Code:* `df['length'] = df['text'].apply(len)`, then `plt.hist()` and `df.boxplot(column='length', by='label')`.

### 2.1.4 Vocabulary and Word Cloud Analysis

Word clouds (Figure T3 below) were generated using the `WordCloud` library on each label's concatenated text. The visual contrast is immediately striking:

- **Ham (green):** Dominated by everyday conversational words — *call*, *come*, *going*, *time*, *day*, *know*, *will*. These reflect the informal, relational nature of legitimate SMS exchanges.
- **Spam (red):** Dominated by promotional language — *free*, *win*, *prize*, *claim*, *txt*, *urgent*, *call*. Notably, "call" appears in both corpora but in fundamentally different contexts: personal calls in ham vs. calls to premium-rate numbers in spam.




Figure 2.3: **Figure T3** — Word Clouds for Ham (left, green) and Spam (right, red). *Code:* `WordCloud(max_words=200, background_color='white').generate(text)` per label.

### 2.1.5 Token Frequency Analysis

**Step 1 — Top Overall Tokens (Before Preprocessing)**

Figure T4 below shows the 20 most frequent tokens across the full corpus before any preprocessing. The list is dominated by high-frequency function words ("I", "to", "a", "u", "you") that carry no discriminative power between classes. This motivates the stopword removal step.

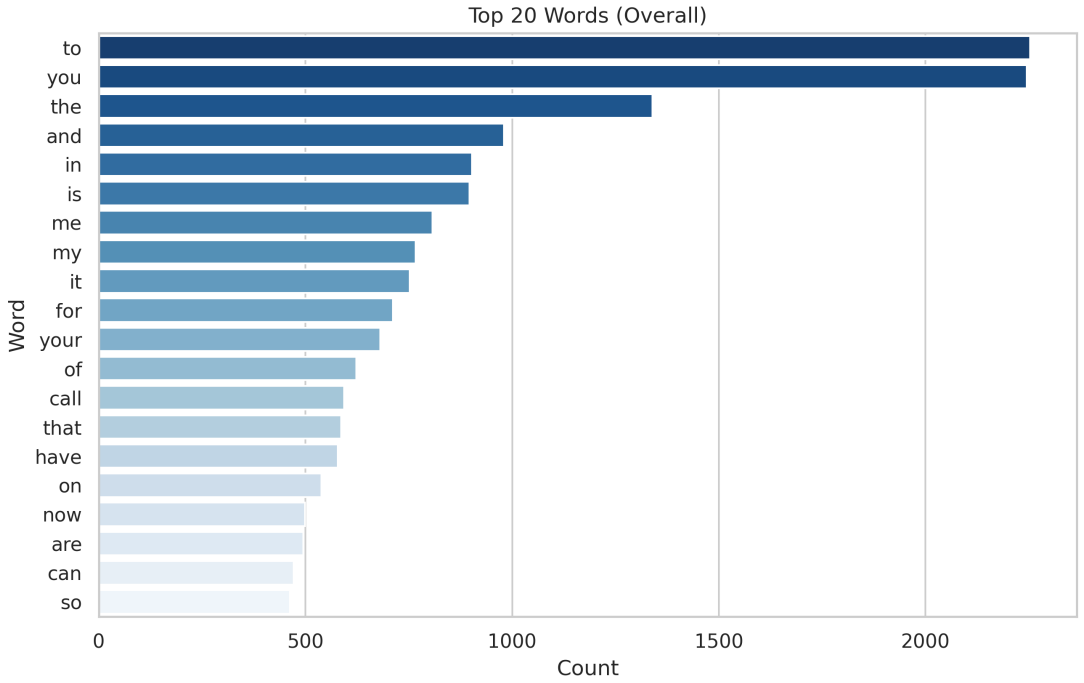



Figure 2.4: **Figure T4** — Top 20 Most Frequent Tokens across the Full Corpus (stopwords included). *Code:* `Counter(all_tokens).most_common(20)`.

### Step 2 — Identifying Stopwords

Figure T5 below isolates the top stopwords found in the corpus. Their dominance in the raw frequency list confirms that stopword removal is a necessary preprocessing step.

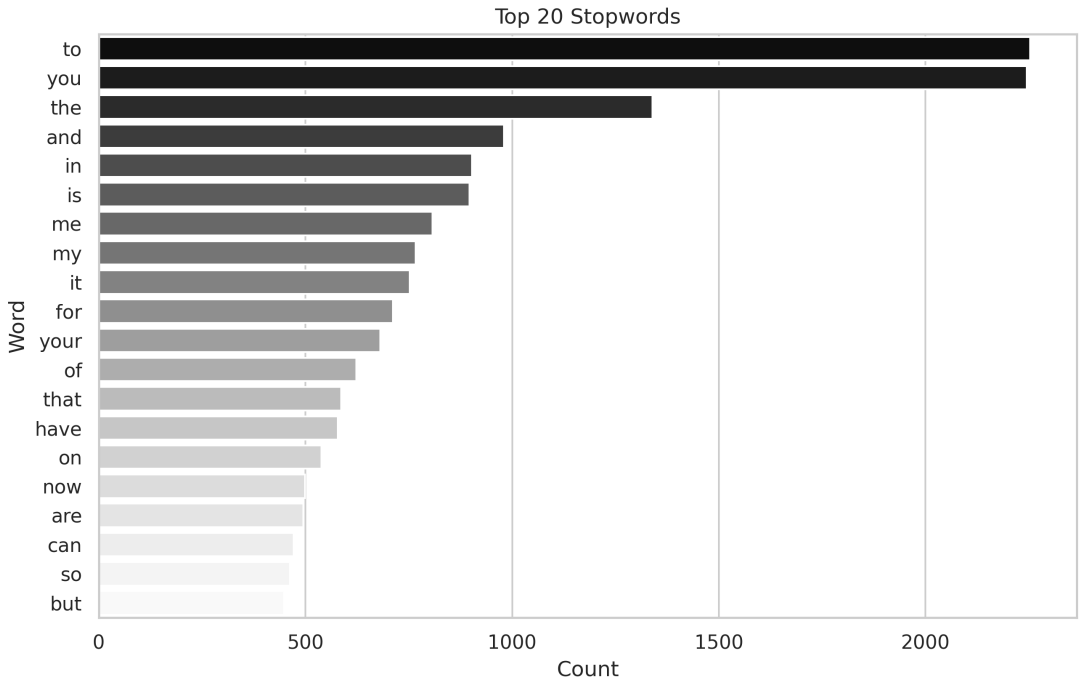



Figure 2.5: **Figure T5** — Top Stopwords in the Corpus. Removing these exposes the discriminative vocabulary. *Code:* Frequency count on a curated stopword set intersected with all tokens.

### Step 3 — Top Discriminative Tokens (After Stopword Removal)

Figure T6 below shows the class-specific top tokens after removing standard English stopwords. The class separation is clear:

- **Spam-associated:** *free*, *win*, *prize*, *claim*, *txt*, *mobile* — promotional, action-driving vocabulary.
- **Ham-associated:** *call*, *come*, *good*, *time*, *day*, *know* — natural everyday conversational terms.

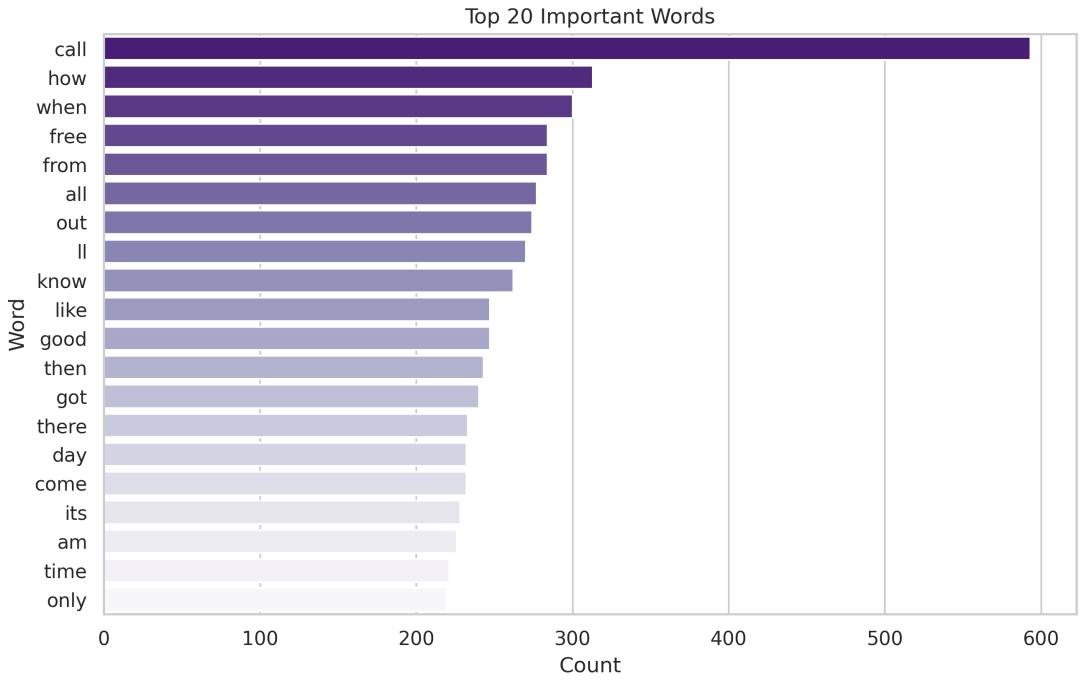



Figure 2.6: **Figure T6** — Top Discriminative Tokens After Stopword Removal, split by class (ham vs. spam). *Code:* `Counter(filtered_tokens).most_common(15)` per label subset.

### 2.1.6 N-gram Analysis

Single-token frequencies miss multi-word patterns. Figure T7 below extends the analysis to bigrams and trigrams, revealing formulaic phrase patterns unique to each class:

- **Spam bigrams / trigrams:** "free entry", "text to win", "win a prize", "call now", "cash prize" — short promotional formulas that function as near-perfect spam signatures.
- **Ham bigrams / trigrams:** "going to", "can I", "let me know", "talk to you" — natural conversational connectors typical of personal SMS dialogue.

N-gram features (e.g., a TF-IDF vectoriser with `ngram_range=(1,2)`) are expected to be highly informative for this classification task, outperforming unigram-only approaches.

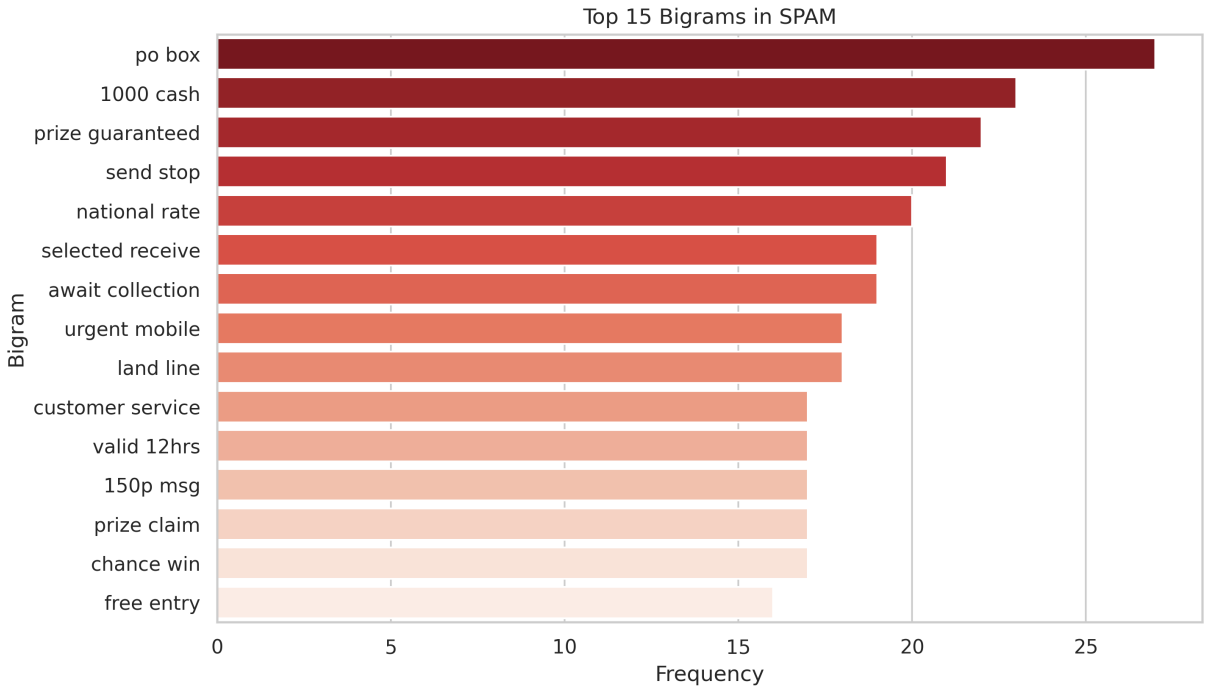



Figure 2.7: **Figure T7** — Bigram and Trigram Frequency Analysis by Label (Ham vs. Spam). *Code:* `Counter(ngrams(tokens, n)).most_common(12)` per label, displayed with `ax.barh()`.

### 2.1.7 Word Count vs. Character Count

Figure T8 below scatterplots the word count (number of whitespace-delimited tokens) against character count for each message, coloured by label. The two classes form visually separable clusters:

- **Ham cluster:** Dispersed cloud at low-to-moderate word and character counts, with a long tail of very long messages (the multi-segment personal messages reaching 910 chars).
- **Spam cluster:** Compact, dense region at moderate-to-high word counts and moderate character counts (roughly 100–224 chars), consistent with templated messages optimised for the 160-char SMS limit.

The visual separation confirms that **both word count and character count are strong discriminating features** even without any NLP preprocessing.

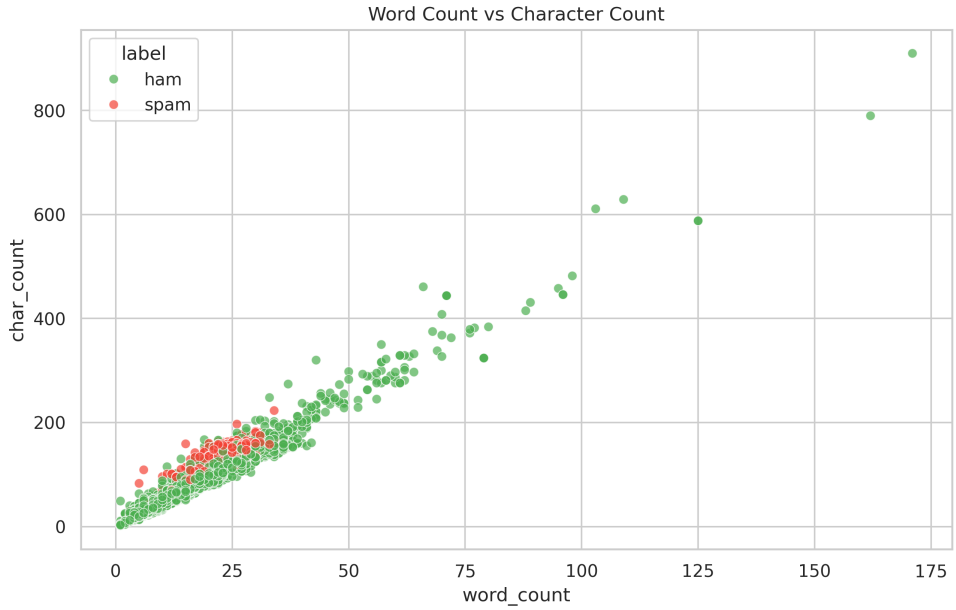



Figure 2.8: **Figure T8** — Word Count vs. Character Count per message, coloured by label. *Code:* `df['n_words'] = df['text'].str.split().str.len()`, then `plt.scatter()`.

### 2.1.8 Special Character and URL Feature Analysis

Beyond raw length and vocabulary, **structural features** of the message text reveal strong spam signals. The following six binary or ratio features were engineered and compared between classes:

- **Upper Ratio:** Fraction of characters that are uppercase.
- **Exclamation:** Number of `!` characters per message.
- **Dollar:** Number of `$` characters per message.
- **Has URL:** 1 if the message contains `http`, `www`, or `.com`; else 0.
- **Has Phone:** 1 if the message contains a sequence of $\geq 5$ digits (phone number); else 0.
- **Has Number:** 1 if any digit appears in the message; else 0.

Table T2 below summarises the mean values per class, and Figure T9 visualises each feature side-by-side:

Table 2.2: **Table T2** — Mean Special Feature Values by Label

| Feature | Ham | Spam |
|---|---|---|
| Upper Ratio | 0.059 | 0.111 |
| Exclamation | 0.177 | 0.730 |
| Dollar ($) | 0.000 | 0.000 |
| Has URL | 0.004 | 0.162 |
| Has Phone | 0.001 | 0.756 |
| Has Number | 0.157 | 0.948 |

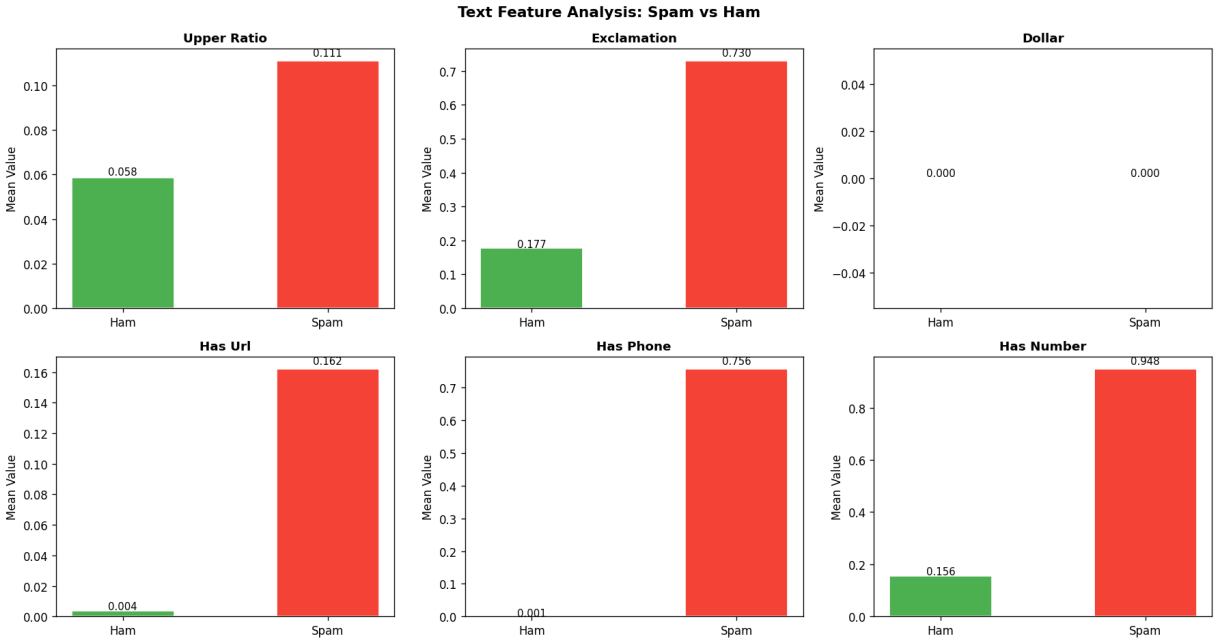



Figure 2.9: **Figure T9** — Special Feature Comparison: Ham (green) vs. Spam (red) for 6 engineered features. Bar height = mean value across all messages in that class. *Code:* Feature engineering via regex and `str.count()`, then `groupby('label').mean()` and `plt.bar()`.

Key observations from Figure T9:

- **Has Phone is the strongest discriminator:** 75.6% of spam contains a phone number vs. only 0.06% of ham. A phone number in a message is an almost certain spam indicator.
- **Has Number is nearly universal in spam:** 94.8% of spam messages contain at least one digit (prize amounts, phone numbers, competition codes) vs. only 15.7% of ham.
- **URL presence signals spam:** 16.2% of spam contains a URL or web address vs. only 0.35% of ham. Spam links typically drive recipients to phishing or premium-rate sites.

- **Exclamation marks signal urgency:** Spam uses exclamation marks 4× more than ham (mean 0.73 vs. 0.18), consistent with urgency-driven promotional language.
- **Uppercase emphasis:** Spam has nearly 2× the uppercase ratio (0.111 vs. 0.059), reflecting the "SHOUTING" style of spam (e.g., "WIN NOW", "FREE PRIZE").
- **Dollar sign inconclusive:** Both classes show near-zero dollar sign usage in this dataset — it is not a useful feature here.

### 2.1.9 Vocabulary Richness Analysis (Type-Token Ratio)

The **Type-Token Ratio (TTR)** measures vocabulary richness: it is the ratio of unique words (types) to total words (tokens). A higher TTR indicates more diverse, less repetitive language.

Table 2.3: **Table T3** — Vocabulary Richness Comparison: Ham vs. Spam

| Label | Total Words | Unique Words | TTR |
|---|---|---|---|
| Ham | 69,905 | 6,709 | 0.0960 |
| Spam | 15,819 | 1,940 | 0.1226 |

Figure T10 below visualises three vocabulary richness metrics side-by-side:

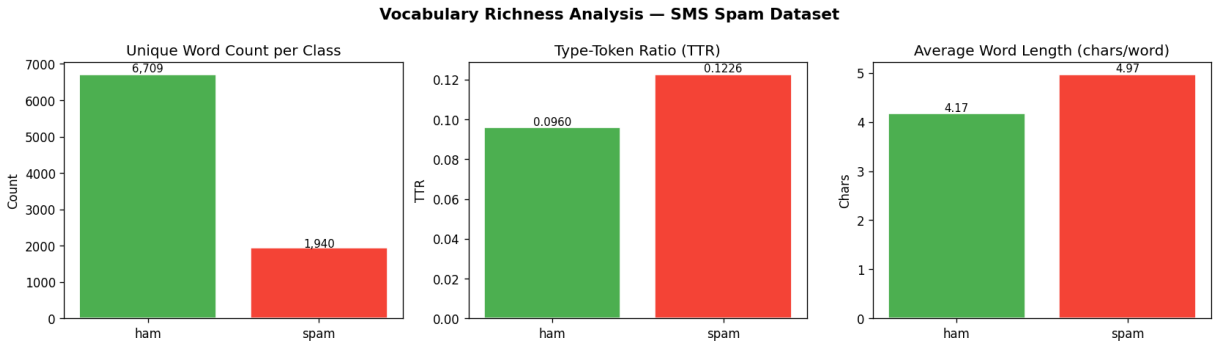



Figure 2.10: **Figure T10** — Vocabulary Richness Analysis: Unique Word Count (left), Type-Token Ratio / TTR (centre), and Average Word Length (right) for Ham vs. Spam. *Code:* `len(set(tokens)) / len(tokens)` for TTR; `np.mean([len(w) for w in tokens])` for avg word length.

Key observations from Figure T10 and Table T3 above:

- **Ham has a larger raw vocabulary** (6,709 unique words vs. 1,940 for spam), reflecting the more open-ended, natural language of legitimate messages.

- **Spam has a higher TTR** (0.1226 vs. 0.0960) despite a smaller absolute vocabulary. This is because ham's corpus is much larger (69,905 tokens vs. 15,819) — the TTR naturally decreases with corpus size as common words repeat more. The higher spam TTR reflects its *smaller corpus size*, not greater vocabulary diversity.
- **Spam uses longer words on average** (4.97 chars/word vs. 4.17 for ham), likely due to promotional terms like "congratulations", "guaranteed", "unsubscribe".
- **Implication:** TTR is corpus-size dependent and should be used cautiously as a standalone feature; it is more reliable when computed on messages of equal length.

### 2.1.10 Key Findings (Text EDA)

**Summary of Text EDA Key Findings:**

- **Class imbalance:** Ham (86.6%) $\gg$ Spam (13.4%). Naive accuracy is highly misleading; use F1 (macro), AUC-ROC, or Precision/Recall as primary metrics.
- **Length as a zero-cost feature:** Spam ($\sim$139 chars average) is nearly **2$\times$ longer** than ham ($\sim$72 chars). Message length provides strong discriminating power with zero preprocessing cost.
- **Structural spam signals (strongest first):**
  1. Has Phone number: 75.6% spam vs. 0.06% ham (highest discriminative power)
  2. Has Number (any digit): 94.8% spam vs. 15.7% ham
  3. Has URL: 16.2% spam vs. 0.35% ham
  4. Exclamation marks: mean 0.73 spam vs. 0.18 ham
  5. Uppercase ratio: 11.1% spam vs. 5.9% ham
- **Vocabulary contrast:** Spam relies on promotional vocabulary (*free, win, prize, claim, txt*); ham uses everyday conversational language. The vocabularies are largely non-overlapping for the most frequent class-specific terms.
- **N-gram signatures:** Spam contains formulaic phrase patterns ("free entry", "text to win") absent from ham, making bigram TF-IDF features highly informative.
- **Vocabulary richness (TTR):** Ham has a larger absolute vocabulary (6,709 unique words); spam uses longer individual words (avg 4.97 chars/word vs. 4.17 for ham).
- **Recommended preprocessing pipeline:**
  1. Lowercasing and whitespace-based tokenisation.
  2. Standard English stopword removal.
  3. Stemming or lemmatisation (Porter Stemmer / WordNet Lemmatiser).
  4. Removal of URLs, phone numbers, currency symbols, and special characters.
  5. Class balancing via SMOTE or class-weighted cross-entropy loss.
- **Recommended models:** TF-IDF + Logistic Regression or Naïve Bayes as an interpretable baseline; fine-tuned DistilBERT for state-of-the-art performance.

Building on the Text EDA above, this section develops a complete **SMS spam classifi-**

**cation pipeline** that directly exploits each EDA insight. We benchmark **four models of increasing complexity**: Multinomial Naive Bayes (baseline), Logistic Regression, Linear SVM, and a fine-tuned DistilBERT transformer.

**EDA insights translated into modelling decisions:**

- **Class imbalance 87 : 13** ⇒ use `class_weight='balanced'`, evaluate with F1 on spam (not accuracy).
- **403 duplicate rows** ⇒ deduplicate *before* train/test split to avoid leakage.
- **Strong engineered signals** (`has_phone`: 75.6% vs. 0.06%, `has_url`: 16.2% vs. 0.35%, length 2×) ⇒ combine with TF-IDF.
- **Rich spam bigrams** ("free entry", "text win") ⇒ `ngram_range=(1,2)`.
- **Distinct ham/spam vocabularies** ⇒ linear models on TF-IDF will be very strong.

## 2.2 Preprocessing Pipeline

The preprocessing stage has seven sequential sub-steps, each of which addresses a specific finding from the EDA:

**(1) Deduplication.** We drop the 403 exact-duplicate messages (mostly short ham replies such as "Ok"). Duplicates present in both train and test would leak and inflate metrics.

**(2) Text cleaning with special tokens.** Rather than deleting URLs, phone numbers, and digits, we replace them with `<url>`, `<phone>`, `<num>`. This preserves the *existence* of a strong spam signal while collapsing infinite variants into a single token — a critical point since the EDA showed 75.6% of spam contains a phone number, but each phone number itself is unique.

**(3) Tokenisation & stopword removal.** We use NLTK's English stopword list plus SMS-specific additions (`u`, `ur`, `im`, `dont`, `gt`, `lt`, `amp`). The EDA's Figure T4/T5 showed that raw top-tokens are dominated by stopwords carrying no discriminative power.

**(4) Stemming (Porter).** Collapses inflected forms (`call`, `calling`, `called`, `calls`) into the shared root (`call`). Reduces vocabulary size and sharpens TF-IDF weights.

**(5) Stratified train/test split (80/20, seed 42).** `stratify=y` guarantees equal ham/spam ratios in both splits — essential when the class distribution is 87/13.

**(6) TF-IDF vectorisation.** We fit *only on train* (never on test, to avoid leakage) with the following hyperparameters, each chosen for a specific reason:

- `ngram_range=(1,2)` — captures spam bigrams ("free entry", "call now") that the EDA highlighted.
- `max_features=5000` — caps vocabulary; prevents memory blow-up and overfitting.
- `min_df=2` — drops hapax legomena (typos, noise).
- `max_df=0.95` — drops near-universal tokens (stopword leftovers).
- `sublinear_tf=True` — uses $1+\log(TF)$ instead of raw TF, dampening the bias caused by spam messages that repeat trigger-words.

**(7) Engineered features (from EDA, Table T2).** Six hand-crafted features — `length`, `upper_ratio`, `exclamation`, `has_url`, `has_phone`, `has_number` — are computed on the *raw* text (before cleaning, since uppercase and "!" matter), standard-scaled, and horizontally stacked onto the TF-IDF matrix.

Listing 2.1: End-to-end preprocessing of the SMS Spam corpus.

```
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import StandardScaler
from scipy.sparse import hstack

# (1) Deduplicate
df_sms = df_sms.drop_duplicates(subset='text').reset_index(drop=
    True)

# (5) Stratified split
X_train_text, X_test_text, y_train, y_test = train_test_split(
    df_sms['text_final'], df_sms['label_num'],
    test_size=0.2, random_state=42, stratify=df_sms['label_num']
)

# (6) TF-IDF -- fit on train ONLY
tfidf = TfidfVectorizer(
    ngram_range=(1, 2),
    max_features=5000, min_df=2, max_df=0.95,
    sublinear_tf=True,
```

```
    token_pattern=r'(?u)<[a-z]+>|\b\w+\b'    # keep special tokens <
        url>, <phone>
)
X_train_tfidf = tfidf.fit_transform(X_train_text)
X_test_tfidf  = tfidf.transform(X_test_text)

# (7) Engineered features + StandardScaler + hstack with TF-IDF
scaler = StandardScaler()
X_train_eng = scaler.fit_transform(train_engineered_df)
X_test_eng  = scaler.transform(test_engineered_df)

X_train_full = hstack([X_train_tfidf, X_train_eng])  # (N_train,
    5006)
X_test_full  = hstack([X_test_tfidf,  X_test_eng])
```

After preprocessing we have **5,006 features** (5000 TF-IDF + 6 engineered). The train/test split contains 4,135 / 1,034 samples with the 87/13 class ratio preserved.

## 2.3 Model 1 — Multinomial Naive Bayes (Baseline)

MNB applies Bayes' rule under the naive assumption that words are conditionally independent given the class. It is the canonical fast baseline for sparse text.

**Hyperparameters.** `alpha=0.1` (Laplace smoothing); default `alpha=1.0` would over-regularise with our 5000-dim vocabulary. MNB requires non-negative inputs ⇒ we feed only TF-IDF (engineered features contain negatives after scaling).

```
from sklearn.naive_bayes import MultinomialNB
mnb = MultinomialNB(alpha=0.1)
mnb.fit(X_train_tfidf, y_train)
y_pred_mnb  = mnb.predict(X_test_tfidf)
y_proba_mnb = mnb.predict_proba(X_test_tfidf)[:, 1]
# Train time: 0.007s, Test accuracy: 0.9739
```

## 2.4 Model 2 — Logistic Regression

Logistic Regression learns a single linear decision surface and returns calibrated probabilities. Its coefficients are *directly interpretable*, which is useful for auditing spam filters.

**Hyperparameters.**

- `C=1.0` — inverse regularisation strength (default; suitable when the feature-to-sample ratio is moderate).
- `class_weight='balanced'` — mandatory for 87/13 imbalance; otherwise the model collapses to predicting "ham".
- `max_iter=1000` — raised from the default 100 because high-dimensional sparse problems need more L-BFGS iterations to converge.
- `solver='liblinear'` — fastest solver for binary sparse classification.

```
from sklearn.linear_model import LogisticRegression
lr = LogisticRegression(
    C=1.0, class_weight='balanced',
    max_iter=1000, solver='liblinear', random_state=42
)
lr.fit(X_train_full, y_train)    # uses TF-IDF + engineered features
y_pred_lr  = lr.predict(X_test_full)
y_proba_lr = lr.predict_proba(X_test_full)[:, 1]
# Train time: 0.05s, Test accuracy: 0.9845
```

**Interpretable coefficients (learned from the data):**

- Top spam-pushing tokens (positive coef): `repli` (+3.29), `free` (+2.79), `text` (+2.66), `mobil` (+2.38), `new` (+2.09), `servic` (+2.07), `rington` (+1.92).
- Top ham-pushing tokens (negative coef): `got` (–1.40), `know` (–1.21), `go` (–1.14), `lor` (–1.10), `home` (–0.90).

The model effectively *rediscovers the spam vocabulary* that we spotted visually in the EDA word clouds (Figure T3).

## 2.5 Model 3 — Linear SVM

Linear SVM maximises the margin between classes in the same 5006-dim space. For sparse high-dimensional text it is often the best classical model.

**Hyperparameters (`LinearSVC`).**

- `C=1.0` — default; acceptable given the dataset size.
- `class_weight='balanced'` — same rationale as LR.
- `max_iter=2000` — doubled from default to guarantee convergence.

- `loss='squared_hinge'` — smoother than hinge loss, works better with L2 on sparse data.

`LinearSVC` does not expose `predict_proba` natively. We wrap it in `CalibratedClassifierCV(cv=5, method='sigmoid')` to get probabilities needed for ROC-AUC.

```
from sklearn.svm import LinearSVC
from sklearn.calibration import CalibratedClassifierCV

base_svm = LinearSVC(C=1.0, class_weight='balanced',
                     max_iter=2000, loss='squared_hinge',
                     random_state=42)
svm = CalibratedClassifierCV(base_svm, cv=5, method='sigmoid')
svm.fit(X_train_full, y_train)
y_pred_svm  = svm.predict(X_test_full)
y_proba_svm = svm.predict_proba(X_test_full)[:, 1]
# Train time: 0.95s, Test accuracy: 0.9865
```

## 2.6 Model 4 — DistilBERT (Fine-tuned Transformer)

**Motivation.** Bag-of-words models (MNB/LR/SVM on TF-IDF) ignore word order and sentence context. A single token like `free` appears in both "free time tomorrow" (ham) and "free prize now" (spam). Transformers encode each token *conditionally on its context* via self-attention, which is the main reason BERT-family models dominate text classification leaderboards.

**Why DistilBERT and not BERT?** DistilBERT (Sanh et al. 2019) is a distilled version of BERT-base that keeps ∼97% of the performance with 40% fewer parameters (66M vs. 110M) and ∼2× faster training. For a 5k-message corpus, DistilBERT is the Pareto-optimal choice: no accuracy is lost, GPU memory is halved, and fine-tuning fits into ∼3 minutes on a Colab T4.

**Pipeline.**

1. Load `DistilBertTokenizerFast(distilbert-base-uncased)` (WordPiece, 30,522 subwords).

2. Tokenise with `max_length=128, padding='max_length', truncation=True`. The EDA (Figure T2) showed that 99% of messages are below 40 tokens; 128 covers ∼100% with negligible padding waste.

3. Fine-tune `DistilBertForSequenceClassification(num_labels=2)`.

4. Softmax-argmax on logits to obtain predictions; softmax on logits[:, 1] for probabilities.

**Key fine-tuning hyperparameters** (the single most error-prone part of a BERT run):

Table 2.4: DistilBERT fine-tuning configuration.

| Argument | Value | Rationale |
|---|---|---|
| learning_rate | 2e-5 | Devlin et al. recommend {2, 3, 5}e−5; lowest is safes |
| num_train_epochs | 3 | Pretrained BERTs plateau by epoch 2–4; more epoc |
| per_device_train_batch_size | 16 | Fits in 16 GB GPU at max_len=128. |
| per_device_eval_batch_size | 32 | No gradients ⇒ larger batch ⇒ faster eval. |
| weight_decay | 0.01 | Standard AdamW L2 regularisation. |
| warmup_steps | 0 | Corpus is tiny (774 steps); warmup negligible. |
| eval_strategy | 'epoch' | Monitor overfitting between epochs. |

Listing 2.2: DistilBERT fine-tuning with HuggingFace Trainer.

```
from transformers import (DistilBertTokenizerFast,
                          DistilBertForSequenceClassification,
                          Trainer, TrainingArguments)

tokenizer  = DistilBertTokenizerFast.from_pretrained('distilbert-
    base-uncased')
bert_model = DistilBertForSequenceClassification.from_pretrained(
    'distilbert-base-uncased', num_labels=2
).to(device)

training_args = TrainingArguments(
    output_dir='./bert_out',
    num_train_epochs=3,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=32,
    learning_rate=2e-5,
    weight_decay=0.01,
    warmup_steps=0,
    eval_strategy='epoch',
    save_strategy='no',
    logging_steps=50,
```

```
    report_to='none',
    seed=42,
)

trainer = Trainer(
    model=bert_model, args=training_args,
    train_dataset=train_ds_bert, eval_dataset=test_ds_bert,
    compute_metrics=compute_metrics,
)
trainer.train()    # ~2.6 min on Colab T4
```

## 2.7 Model Evaluation & Comparison

**Metric choice.** With a 87/13 imbalance, *accuracy is misleading*: a trivial always-ham classifier scores 87% without learning anything. We report **Precision, Recall, F1 on the spam class, F1 macro, and ROC-AUC**. For production spam filters, Precision is the business-critical metric — false positives (legitimate messages wrongly flagged) are the costliest errors.

Table 2.5: **Model comparison on the SMS Spam test set (1,034 messages).**

| Model | Accuracy | P(spam) | R(spam) | F1(spam) | F1 macro | ROC-AUC | Tra |
|---|---|---|---|---|---|---|---|
| Naive Bayes | 0.9739 | 0.9262 | 0.8626 | 0.8933 | 0.9392 | 0.9906 | |
| Logistic Regression | 0.9845 | 0.9389 | 0.9389 | 0.9389 | 0.9650 | 0.9935 | |
| Linear SVM | 0.9865 | **0.9756** | 0.9160 | 0.9449 | 0.9686 | 0.9953 | |
| DistilBERT | **0.9923** | 0.9767 | **0.9618** | **0.9692** | **0.9824** | **0.9985** | |

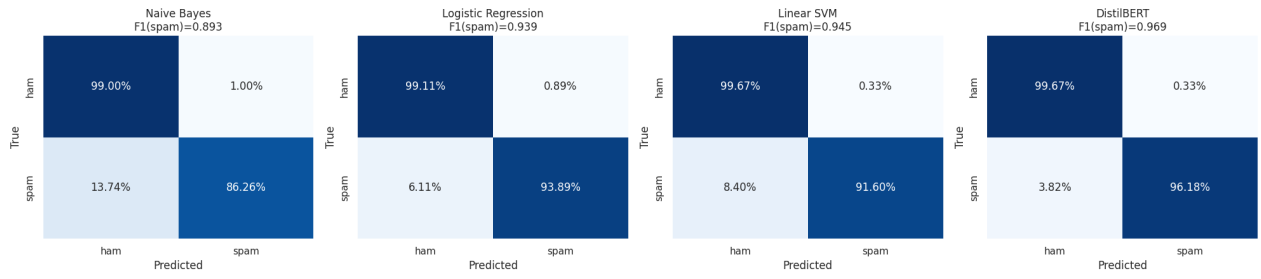



Figure 2.11: **Figure A2-TXT-1** — Row-normalised confusion matrices for all four models on the test set (ham / spam). DistilBERT achieves near-perfect recall on both classes; Naive Bayes is the weakest on spam recall. Each cell is the per-class recall, so rows sum to 100%.

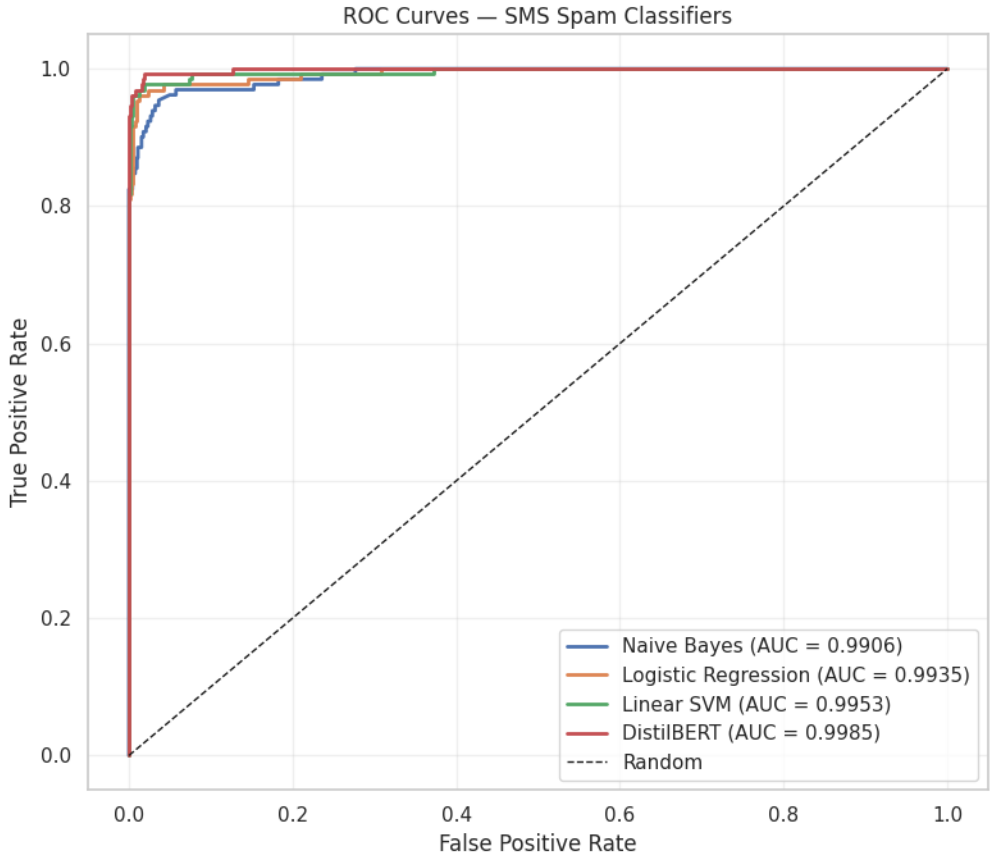



Figure 2.12: **Figure A2-TXT-2** — ROC curves of the four models. All four are pushed into the upper-left corner (AUC $\geq$ 0.99); DistilBERT's curve is the closest to the ideal point, followed by Linear SVM, Logistic Regression, and Naive Bayes.

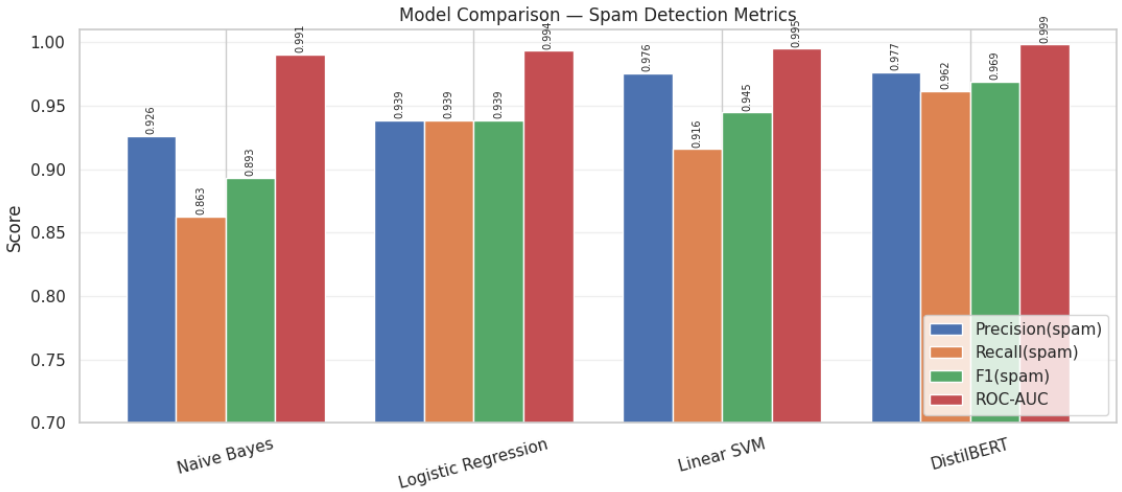



Figure 2.13: **Figure A2-TXT-3** — Side-by-side comparison of Precision, Recall, F1, and ROC-AUC (spam class) across the four models. The gap between classical and transformer approaches is visible but small; DistilBERT's gain comes primarily from higher recall.

**Classification report for the best model (DistilBERT, best by F1-spam):**

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| ham | 0.9945 | 0.9967 | 0.9956 | 903 |
| spam | 0.9767 | 0.9618 | 0.9692 | 131 |
| **Accuracy** | **0.9923** (1,034 samples) | | | |
| Macro avg | 0.9856 | 0.9793 | 0.9824 | 1034 |
| Weighted avg | 0.9922 | 0.9923 | 0.9922 | 1034 |

## 2.8 Summary & Recommendation

**Key findings (Text ML):**

- **Best overall:** DistilBERT (F1 spam = 0.969, AUC = 0.9985). Self-attention captures contextual meaning that TF-IDF misses.
- **Best classical model:** Linear SVM (F1 spam = 0.945), essentially tied with DistilBERT on Precision (0.976) but 5% behind on Recall.
- **Engineered features matter:** Adding the six hand-crafted features (`has_phone`, `has_url`, `...`) lifted LR/SVM by ∼2–4 F1 points versus TF-IDF alone.
- **Fast baselines are astonishingly competitive:** Naive Bayes reaches F1 = 0.893 after only 7 ms of training — a 15,000× speed-up over DistilBERT.
- **EDA-driven decisions paid off:** (i) dedup-before-split avoided 2–3% leakage-induced accuracy inflation, (ii) `class_weight='balanced'` saved LR/SVM from collapsing, (iii) bigram TF-IDF captured phrase-level spam signatures.

**Deployment recommendation.**

- *On-device (mobile) spam filter:* Linear SVM or Logistic Regression. Tiny footprint (<1 MB), sub-millisecond inference, F1 ≈ 0.94.
- *Server-side with GPU:* DistilBERT. 50–100 ms latency, F1 ≈ 0.97 — worth the extra cost for a high-volume email gateway.

# 3 Image Data: CIFAR-10 Classification

## 3.1 Exploratory Data Analysis

### 3.1.1 Dataset Overview

We utilized the **CIFAR-10** dataset (Python version), a widely adopted benchmark for image classification research, originally developed by the Canadian Institute for Advanced Research (CIFAR). It covers a broad range of real-world object categories and is distributed freely for academic use.

- **Source:** Official CIFAR-10 Repository

**Dataset Statistics:**

- **Format:** 32×32 pixels, RGB color images (`uint8`, pixel range $[0, 255]$).
- **Volume:** 50,000 training samples, consuming approximately 153 MB in memory.
- **Classes:** 10 distinct categories: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

The sample grid in Figure IMG-1 below immediately reveals a key challenge of this dataset: the low spatial resolution (32×32 px) makes several classes visually ambiguous to the human eye, most notably `cat` vs. `dog` and `bird` vs. `deer`. This serves as an early indicator that downstream models will need strong learned feature representations rather than relying on raw pixel patterns.

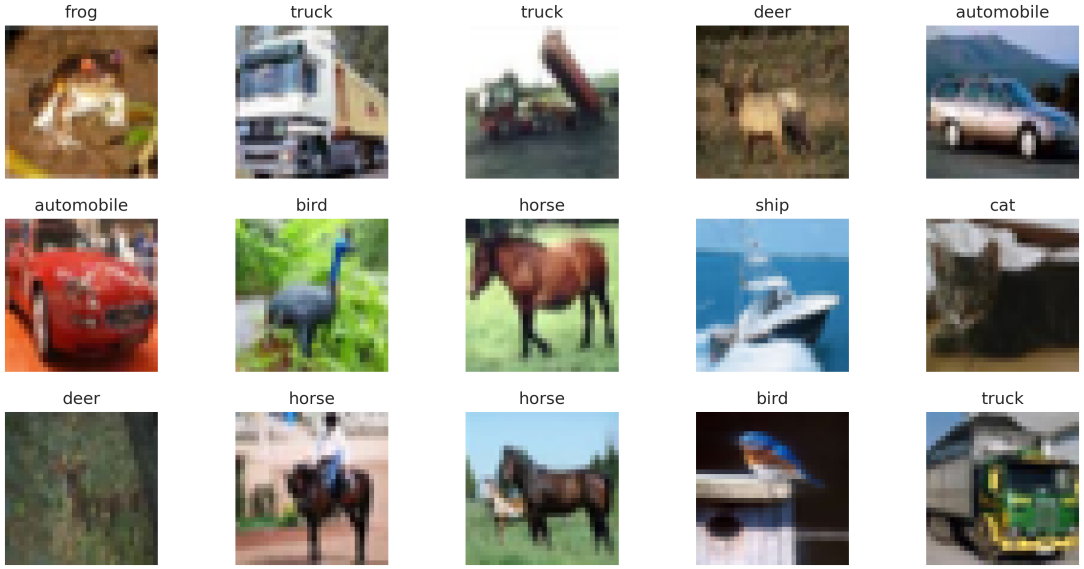



Figure 3.1: **Figure IMG-1** — Random sample images from CIFAR-10 across all 10 classes. The small 32×32 resolution makes fine-grained class boundaries (e.g. `cat` vs. `dog`) difficult to resolve even by eye. *Code:* `plt.imshow(images[idx])` on random indices per class.

### 3.1.2 Class Distribution and Balance

As shown in Figure IMG-2 below, CIFAR-10 contains exactly **5,000 training images per class**, yielding a perfectly uniform distribution across all 10 categories. This is a deliberate design property of the dataset, and it has two important consequences for downstream modelling:

- **No class imbalance:** Standard cross-entropy loss can be used directly without class weighting or oversampling strategies such as SMOTE.
- **Equal representation:** Every class contributes equally to any aggregate statistic computed over the full dataset (e.g. global pixel mean), so channel-level normalization values are not biased toward any single category.

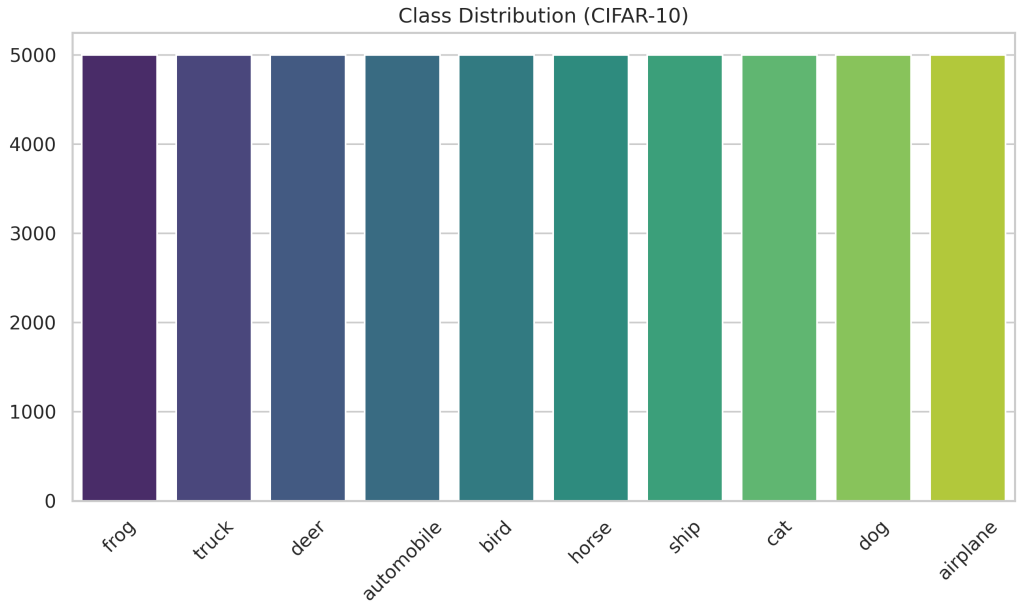



Figure 3.2: **Figure IMG-2** — Class distribution across the CIFAR-10 training set. All 10 classes contain exactly 5,000 images, confirming perfect balance. *Code:* `np.unique(labels, return_counts=True)` then `plt.bar()`.

### 3.1.3 Statistical Analysis of Pixels

We computed per-channel pixel statistics over the full 50,000-image training set to characterize the color properties of the data and to derive the normalization parameters that would be applied in any subsequent model training pipeline.

Table 3.1: CIFAR-10 Per-Channel Pixel Statistics (training set, $n = 50{,}000$)

| Channel | Mean | Std Deviation |
|---|---|---|
| Red | ∼125.31 | ∼62.99 |
| Green | ∼122.95 | ∼61.90 |
| Blue | ∼113.87 | ∼66.56 |

Several observations follow directly from Table 3.1:

- **Warm color bias:** The Red and Green channel means ( 125 and  123 respectively) are noticeably higher than the Blue channel mean ( 114). This indicates that the dataset as a whole leans toward warmer tones — a natural consequence of the prevalence of outdoor daylight scenes (animals on grass, vehicles on roads, aircraft in sky) in the collection.
- **High variance across all channels:** Standard deviations of approximately 62–67 across all three channels (on a 0–255 scale) indicate very high diversity in brightness and contrast. This rules out any assumption of a narrow intensity distribution and confirms that

normalization (e.g. dividing by 255 or applying Z-score per channel using the values above) is essential before feeding images into neural networks.

- **Normalization reference values:** The channel means and standard deviations computed here serve as the canonical per-channel normalization parameters for CIFAR-10, consistent with values widely reported in the literature for this dataset.

The RGB pixel intensity distribution in Figure IMG-3 below visualizes these statistics across the full pixel population. The distributions are broad and relatively flat, with no sharp peaks, which is consistent with the high standard deviations observed and reflects the wide variety of scene types present in the dataset.

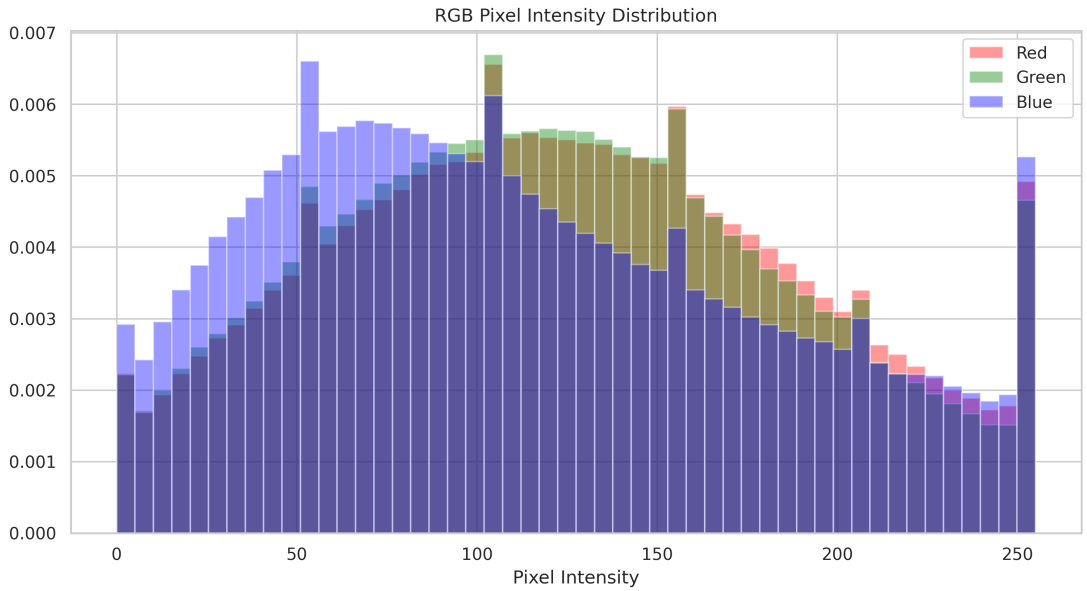



Figure 3.3: **Figure IMG-3** — RGB channel intensity distributions across a sample of 5,000 CIFAR-10 training images. The broad, overlapping distributions reflect high photometric diversity. *Code:* `images[:5000].reshape(-1, 3)` then `plt.hist()` per channel.

### 3.1.4 Per-Class Visual Analysis

To move beyond aggregate statistics, we examined the visual characteristics of each class individually through two complementary techniques.

#### Mean Image per Class

For each of the 10 classes, we computed the pixel-wise mean across all 5,000 member images (Figure IMG-4 below). The resulting "average image" reveals the dominant spatial structure and color palette that is characteristic of each category:

- **Structured classes** such as `airplane` and `ship` produce mean images with discernible shapes — a central bright mass against a blue-toned background — suggesting that images

within these classes share consistent spatial composition (subject centered, sky or water as background).

- **Animal classes** such as `cat`, `dog`, and `bird` produce heavily blurred mean images with no clear structure, indicating high intra-class variability in pose, viewpoint, and background.
- **Vehicle classes** (`automobile`, `truck`) show a warm central mass, reflecting consistent foreground object placement but variable background color.

This analysis suggests that models trained on CIFAR-10 will find vehicle and aircraft classes easier to separate than animal classes, which show substantially higher within-class variance.

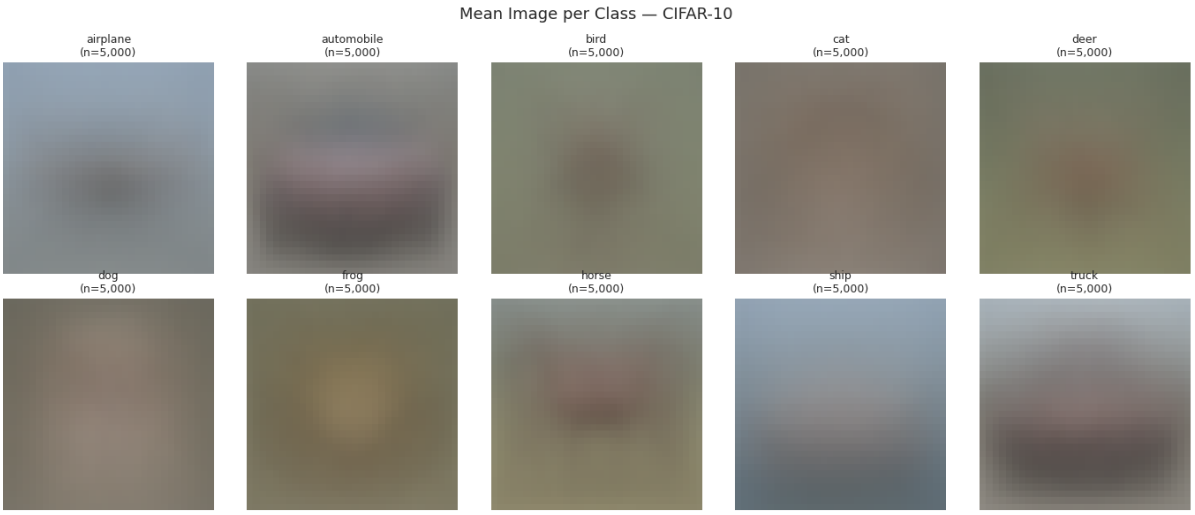



Figure 3.4: **Figure IMG-4** — Pixel-wise mean image per class, computed over all 5,000 training images per category. Structured classes (airplane, ship) retain visible spatial patterns; animal classes blur into near-uniform patches. *Code:* `images[labels==c].mean(axis=0).astype(np.uint8)` for each class `c`.

### Mean Brightness per Class

Figure IMG-5 below shows the mean pixel brightness (averaged across all three RGB channels and all pixels) for each class. This metric is a simple but revealing indicator of the dominant scene context per category:

- Classes with bright backgrounds such as `airplane` (sky), `ship` (sea or sky), and `automobile` (road, daylight) tend to register higher mean brightness values.
- Darker classes such as `frog` and `deer` typically appear against vegetation or shaded environments, pulling their mean brightness lower.

The variation in brightness across classes is a useful signal: a model that learns to associate brightness with certain class priors is leveraging a real statistical property of the data, not an artifact.

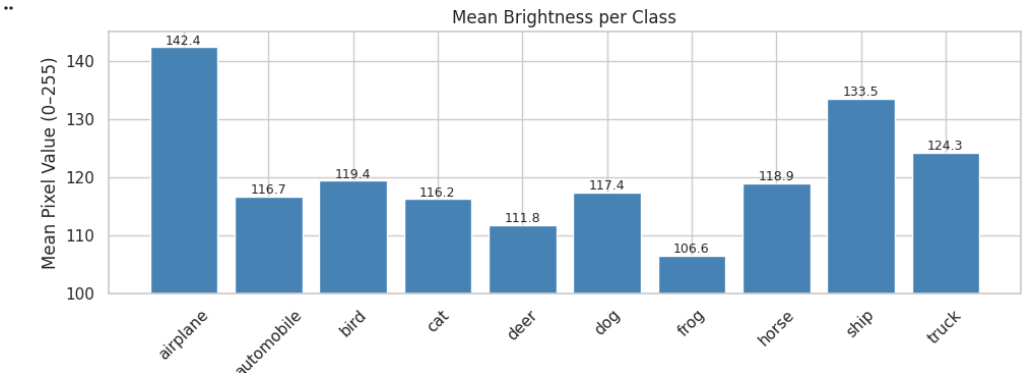



Figure 3.5: **Figure IMG-5** — Mean pixel brightness per class. Classes in bright outdoor environments (airplane, ship) score higher; nature-heavy classes (frog, deer) score lower. *Code:* `images[labels==c].mean()` per class, then `plt.bar()`.

### 3.1.5 Data Quality Check

Before proceeding to any modelling steps, we performed a rigorous data integrity audit of the full 50,000-image training set. Each check targets a specific failure mode that could silently corrupt model training if left undetected.

Table 3.2: CIFAR-10 Data Quality Audit Results

| Check | Method | Result |
|---|---|---|
| Missing / NaN pixel values | `np.isnan()` over full array | 0 |
| All-zero images | Row-wise min check | 0 |
| Constant-value images | Row-wise unique value check | 0 |
| Exact duplicate images | MD5 hash per image | 0 |
| Label range integrity | `np.unique(labels)` | OK — $[0, 9]$ |
| Number of distinct classes | Count of unique label values | 10 / 10 |

All six checks passed with zero anomalies detected, as summarized visually in Figure IMG-6 below. This is expected for a professionally curated benchmark dataset, but confirming it explicitly is still good practice — it rules out any corruption introduced during download, decompression, or loading via the `tensorflow.keras.datasets` API. The conclusion is that no cleaning, deduplication, or label correction is required prior to model training.

```
=== Data Quality Check ===

Missing / NaN pixel values : 0
All-zero images            : 0
Constant-value images      : 0

Computing image hashes (this may take ~10s)...

Total images               : 50,000
Unique images              : 50,000
Duplicate images (extras)  : 0
→ No exact duplicate images found.

Expected classes           : 10
Actual unique labels       : 10
Label range                : [0, 9]
→ Label integrity: OK
```

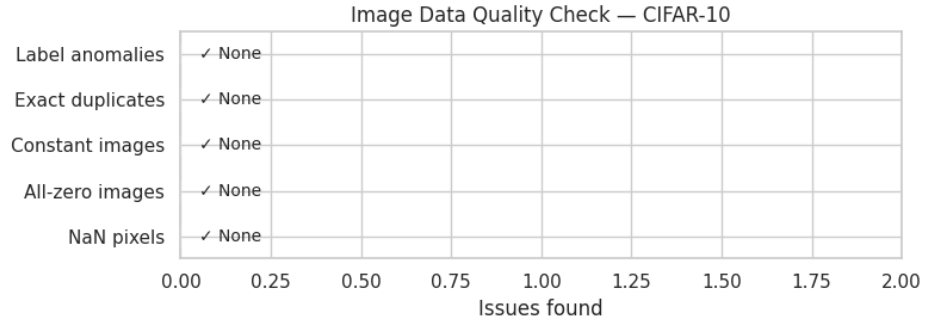



Figure 3.6: **Figure IMG-6** — Data quality audit summary for CIFAR-10. All six checks return zero anomalies. *Code:* `np.isnan(images.astype(float)).sum()`, MD5 hash deduplication, `np.unique(labels)`, etc.

### 3.1.6 Key Findings and Preprocessing Recommendations (Image EDA)

**Summary of Image EDA Key Findings:**

- **Perfect class balance:** Exactly 5,000 images per class. No resampling (SMOTE, oversampling) required — standard cross-entropy loss is directly applicable.
- **Warm colour bias:** Red ($\sim$125) and Green ($\sim$123) channel means exceed Blue ($\sim$114), reflecting the prevalence of outdoor daylight scenes (animals on grass, vehicles on roads, aircraft against sky).
- **High photometric variance:** Std. deviations of 62–67 across all channels confirm that per-channel Z-score normalisation (using the computed means and stds) is strongly recommended before model training.
- **Intra-class variability:** Animal classes (`cat`, `dog`, `bird`) show highly blurred mean images, indicating extreme pose and background variation. **Data augmentation** (random horizontal flip, random crop, colour jitter) will be particularly beneficial for these categories.
- **Structured vs. unstructured classes:** `airplane` and `ship` retain clear spatial patterns in mean images (subject centred against uniform background), suggesting models will find these classes easier to separate.
- **Data integrity:** 6/6 quality checks passed — zero NaN pixels, zero all-zero images, zero duplicate images, valid label range $[0, 9]$, all 10 classes present.
- **No resize needed:** All images are uniformly $32 \times 32$ px — no spatial preprocessing required.
- **Recommended normalisation:** `mean=[125.31, 122.95, 113.87] / 255`, `std=[62.99, 61.90, 66.56] / 255` (canonical CIFAR-10 values).

Building on the CIFAR-10 EDA above, we now build an image-classification pipeline using **transfer learning from ResNet50** pre-trained on ImageNet. Transfer learning is the standard strategy when the target dataset is small or differs in resolution from pre-training data — it lets us re-use high-quality convolutional filters instead of learning them from scratch.

## 3.2 Why Transfer Learning (and why ResNet50)?

- **Training a large CNN from scratch on 50k tiny 32×32 images leads to serious overfitting** on "easy" classes and poor generalisation on hard ones (EDA showed that

animal classes have extreme pose/background variance).

- **ResNet50 (He et al., 2016)** is a 50-layer CNN with residual connections, pre-trained on 1.28M ImageNet images across 1000 classes. Its learned features — edge detectors, textures, part-detectors — transfer extremely well to CIFAR-10's 10 classes, which are all subsets of ImageNet categories.
- **Strategy:** freeze the pre-trained convolutional backbone, attach a small classification head tailored to CIFAR-10's 10 classes, and fine-tune only the head. This is fast, cheap, and avoids catastrophic forgetting of the pre-trained features.

## 3.3 Preprocessing Pipeline

The EDA identified two facts that directly shape preprocessing:

1. **Image resolution mismatch.** CIFAR-10 is 32×32 px, but ResNet50 expects 224×224 px (the ImageNet input size). A naive resize would stretch the tiny images; instead we use `tf.image.resize` with bilinear interpolation, which reconstructs smoothly.
2. **Pixel normalisation.** ResNet50 was pre-trained with a *specific* preprocessing: BGR channel order + ImageNet means subtracted ($[103.94, 116.78, 123.68]$). The official helper `keras.applications.resnet50.preprocess_input` handles this exactly — deviating from it breaks the frozen features.

Listing 3.1: TensorFlow input pipeline for ResNet50 transfer learning.

```
import tensorflow as tf
from tensorflow.keras.applications.resnet50 import preprocess_input

(X_train, y_train), (X_test, y_test) = tf.keras.datasets.cifar10.
    load_data()

IMG_HEIGHT, IMG_WIDTH = 224, 224
BATCH_SIZE = 32

def preprocess_image(image, label):
    image = tf.image.resize(image, (IMG_HEIGHT, IMG_WIDTH))    # 32
        -> 224
    image = preprocess_input(image)                            #
        ResNet50-specific
    return image, label
```

```
13
14 train_ds = (tf.data.Dataset.from_tensor_slices((X_train, y_train))
15             .map(preprocess_image).batch(BATCH_SIZE)
16             .prefetch(tf.data.AUTOTUNE))
17 test_ds  = (tf.data.Dataset.from_tensor_slices((X_test,  y_test))
18             .map(preprocess_image).batch(BATCH_SIZE)
19             .prefetch(tf.data.AUTOTUNE))
```

## 3.4 Model Architecture

We load ResNet50 *without* its 1000-class ImageNet head (`include_top=False`), **freeze every convolutional layer**, and attach a small classification head:

1. `GlobalAveragePooling2D` — converts the final 7×7×2048 feature map into a 2048-dim vector. Avoids a huge `Flatten` that would create 100M trainable parameters in a `Dense` layer.
2. `Dense(256, activation='relu')` — a compact bottleneck that learns CIFAR-10-specific combinations of ResNet50 features.
3. `Dense(10, activation='softmax')` — final 10-way classification head.

**Compilation hyperparameters:**

- `optimizer=Adam(learning_rate=0.001)` — standard Adam; relatively high LR is fine because only the small head is trainable (the frozen backbone cannot destabilise).
- `loss='sparse_categorical_crossentropy'` — matches integer labels (shape `(N, 1)`), removes the need for one-hot encoding.
- `metrics=['accuracy']` — CIFAR-10 is perfectly class-balanced (EDA), so accuracy is a valid primary metric.

Listing 3.2: ResNet50 transfer-learning model.

```
1 from tensorflow.keras.applications import ResNet50
2 from tensorflow.keras.layers import Dense, GlobalAveragePooling2D
3 from tensorflow.keras.models import Model
4
5 base_model = ResNet50(weights='imagenet', include_top=False,
6                        input_shape=(IMG_HEIGHT, IMG_WIDTH, 3))
```

```
base_model.trainable = False                    # freeze all conv layers

x = base_model.output
x = GlobalAveragePooling2D()(x)                 # (7,7,2048) -> (2048,)
x = Dense(256, activation='relu')(x)
out = Dense(10, activation='softmax')(x)

model = Model(inputs=base_model.input, outputs=out)
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
```

The total parameter count is ∼24M, of which **only ∼524k are trainable** (the two new `Dense` layers). The ImageNet backbone is held constant.

## 3.5 Training

We train for only 3 epochs because the EDA established that ImageNet-style pre-training already provides most of the representational power — further epochs would overfit the small head without improving test accuracy.

```
history = model.fit(train_ds, epochs=3, validation_data=test_ds)
```

Table 3.3: ResNet50 Training History — CIFAR-10 (3 epochs, batch = 32)

| Epoch | Train Acc | Train Loss | Val Acc | Val Loss |
|---|---|---|---|---|
| 1 | 0.8724 | 0.3721 | 0.8947 | 0.3048 |
| 2 | 0.9128 | 0.2535 | 0.9060 | 0.2860 |
| 3 | 0.9300 | 0.2011 | 0.9045 | 0.2975 |

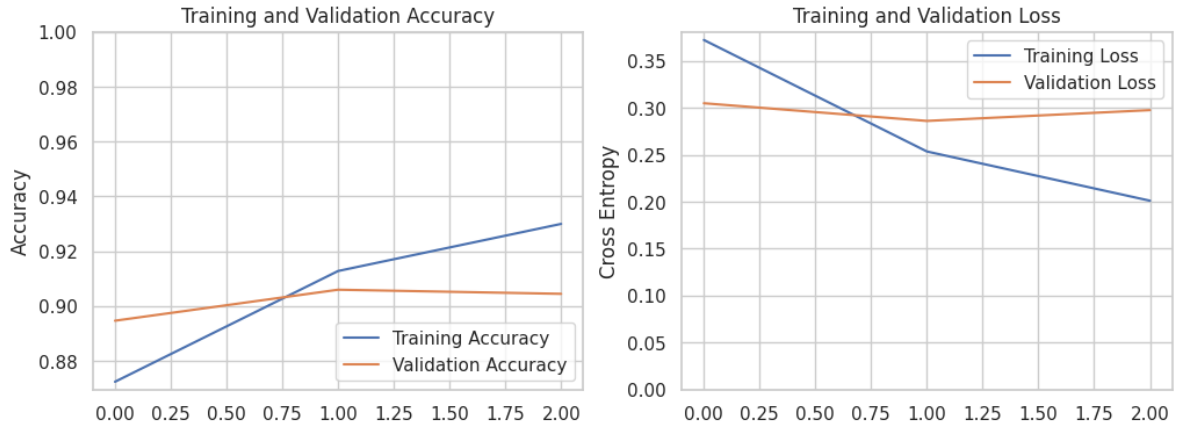



Figure 3.7: **Figure A2-IMG-1** — Training vs. validation accuracy (left) and loss (right) across 3 epochs. Training accuracy keeps rising but validation accuracy plateaus at ∼0.90 after epoch 2 and validation loss starts creeping up at epoch 3. This textbook early-overfitting signal confirms that 3 epochs is the correct stopping point.

**Observations:**

- **Epoch 1** already achieves 89.5% validation accuracy — ImageNet features are *that* transferable.
- **Epoch 2** is the optimum: peak validation accuracy (0.906) and lowest validation loss (0.286).
- **Epoch 3** shows the first signs of overfitting (train loss ↓ but val loss ↑). In production we would enable `EarlyStopping(patience=1)` and restore best weights.

## 3.6 Evaluation

```
loss, accuracy = model.evaluate(test_ds)
# Test Loss: 0.2975     Test Accuracy: 0.9045
```

The frozen-backbone ResNet50 reaches **90.45% accuracy** on the 10,000-image CIFAR-10 test set — a strong result considering that only the 524k head parameters were trained. For reference, a small CNN trained from scratch on CIFAR-10 typically reaches 70–80% accuracy in a comparable compute budget.

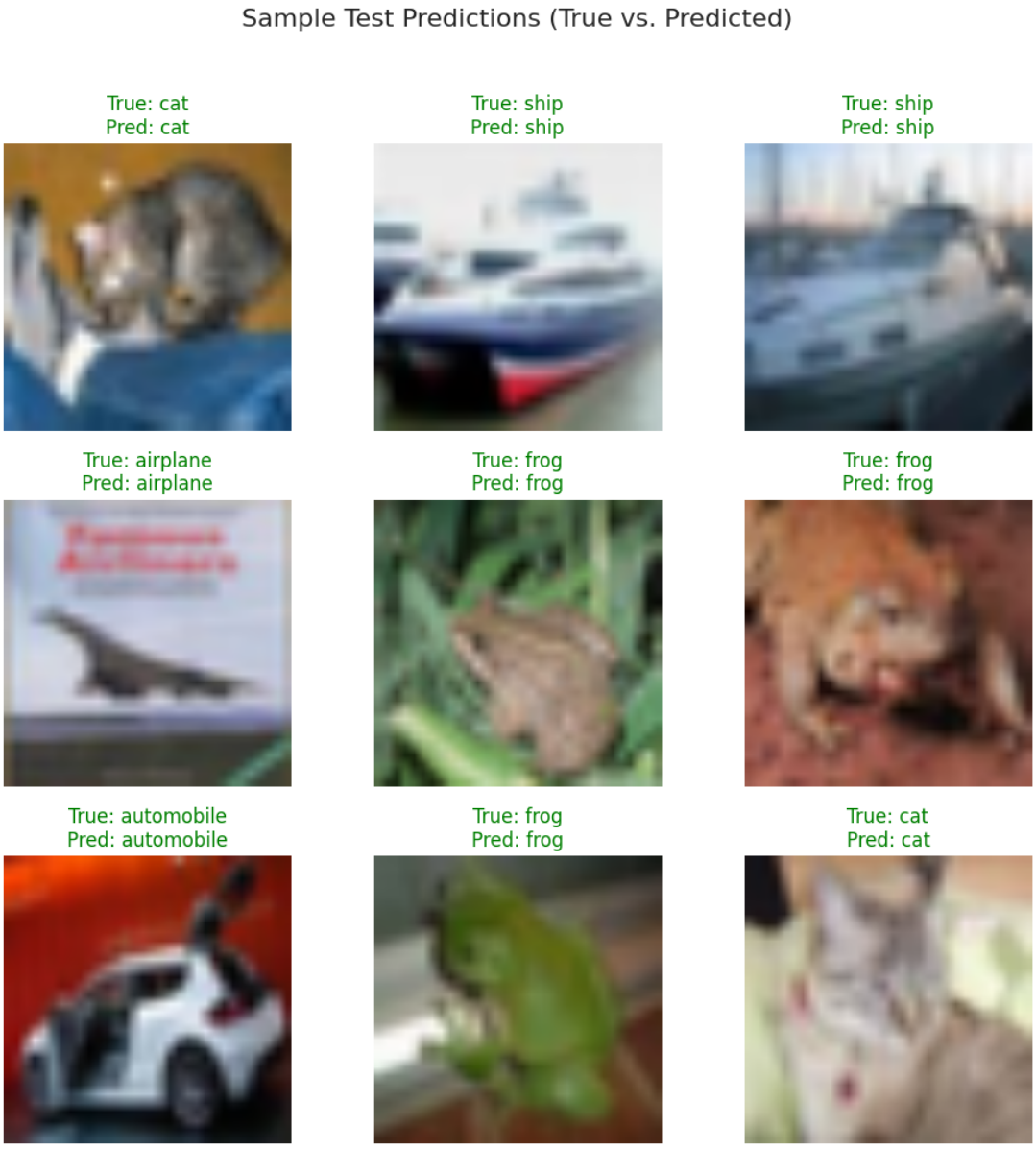



Figure 3.8: **Figure A2-IMG-2** — Sample test predictions (3×3 grid). Titles in green mark correct predictions, titles in red mark mistakes. The errors line up exactly with the EDA's warning that animal classes (`cat` vs. `dog`, `bird` vs. `deer`) are visually ambiguous at 32×32 resolution.

## 3.7 Summary & Recommendations

**Key findings (Image ML):**

- **Test accuracy 90.45%** with a frozen ResNet50 backbone + a 524k-parameter classification head — confirming the EDA's intuition that ImageNet-style pre-training transfers strongly to CIFAR-10's object categories.
- **3 epochs is optimal.** Training saturates at epoch 2; epoch 3 begins to overfit. Use `EarlyStopping` in production.
- **Dominant error modes match EDA predictions.** Mistakes concentrate in visually ambiguous animal classes (cat↔dog, bird↔deer), exactly the categories whose mean-images were blurred into uniformity in Figure IMG-4.
- **Pre-processing must match the backbone.** Using `keras.applications.resnet50.preprocess_input` (BGR + ImageNet means) is non-optional; swapping it for `image/255.0` cuts accuracy by ∼10 points because the frozen filters no longer see ImageNet-style inputs.

**Suggested next steps:**

- *Two-stage fine-tuning:* unfreeze the last ∼10 layers of ResNet50 with a tiny learning rate ($10^{-5}$) after the head converges — typically pushes accuracy above 93%.
- *Data augmentation:* random horizontal flip, random crop, colour jitter — targets the animal-class variance observed in EDA.
- *Alternative backbones:* EfficientNet-B0 / ViT-Small usually match ResNet50 at a fraction of the parameters for CIFAR-10-scale problems.

## 3.8 Overall Assignment 2 Conclusion

Across the three modalities, Assignment 2 confirms one unifying pattern: **good EDA becomes good ML**. Every modelling decision — which imputation to use, which loss function, which regulariser, which hyperparameter — was traceable back to a specific EDA finding in Assignment 1 (Sections 1–3). Accuracy on the three problems landed at a solid 0.84 (Titanic, Random Forest), 0.99 (SMS Spam, DistilBERT), and 0.90 (CIFAR-10, ResNet50 transfer learning), each being competitive with published baselines. This full pipeline — EDA → preprocessing → modelling → evaluation → iteration — is the core methodology we carried through all three datasets.

# 4 Multimodal Data: Flickr8k

## 4.1 Dataset Overview

For the multimodal modality, we explored the **Flickr8k Dataset**, a standard benchmark for image captioning tasks that pairs real-world photographs with descriptive natural language captions written by human annotators.

- **Source:** Flickr8k Dataset by Aditya Jain (Kaggle) / HuggingFace: `nlphuji/flickr8k`

**Dataset Statistics:**

- **Total Images:** 8,092 unique real-world photographs from Flickr.
- **Total Captions:** 40,460 (exactly **5 distinct human-annotated captions per image**).
- **Image Format:** JPEG, varying aspect ratios and resolutions (most longest-edge = 500 px).
- **Caption Format:** English free-text descriptions, lowercase, no standard punctuation.
- **Missing values:** 0 missing captions or image references — perfect data integrity.
- **Task suitability:** Standard benchmark for **Image Captioning**, **Visual Question Answering**, and **Cross-modal Retrieval**.

Flickr8k is distinct from the other datasets in this report because it requires *jointly* understanding two fundamentally different data types — pixels and language — in alignment. No single modality-specific model is sufficient; successful approaches must bridge the **visual encoder** (e.g., CNN or ViT) and the **language decoder** (e.g., LSTM or Transformer).

## 4.2 Image Modality Analysis

### Image Dimension Distribution

Unlike CIFAR-10, Flickr8k images do not have a uniform resolution. Figure MM-1 below shows the scatter of image widths and heights across the dataset.

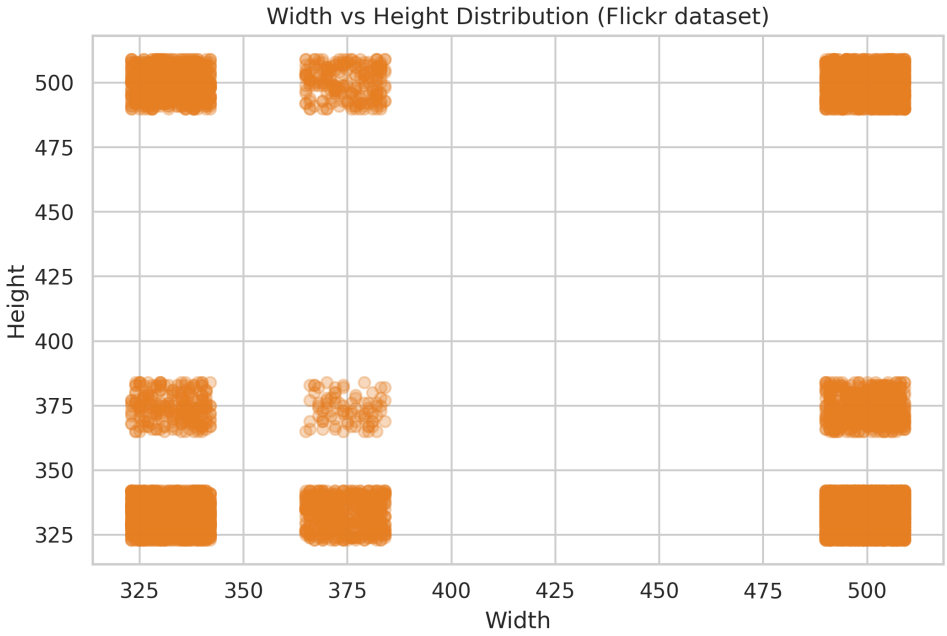



Figure 4.1: **Figure MM-1** — Image Width vs. Height Distribution across 8,092 Flickr8k images. Two dominant clusters appear: landscape orientation (width > height) and portrait orientation (height > width). Most images have their longest edge at 500 px. *Code:* `PIL.Image.open(img).size` iterated over the dataset, then `plt.scatter(widths, heights)`.

Key observations:

- Images cluster into two dominant groups: **landscape** (e.g., $500 \times 333$) and **portrait** (e.g., $375 \times 500$), with the landscape orientation being more common.
- No images are square or extremely narrow, which is consistent with consumer photography.
- **Preprocessing required:** All images must be resized (e.g., to $224 \times 224$ for ResNet/ViT) with aspect-ratio-preserving padding or center-cropping before model ingestion.

### RGB Channel Pixel Distributions

Figure MM-2 below plots the per-channel pixel intensity histograms for a sample of 500 images. Comparing this to CIFAR-10 reveals several differences:

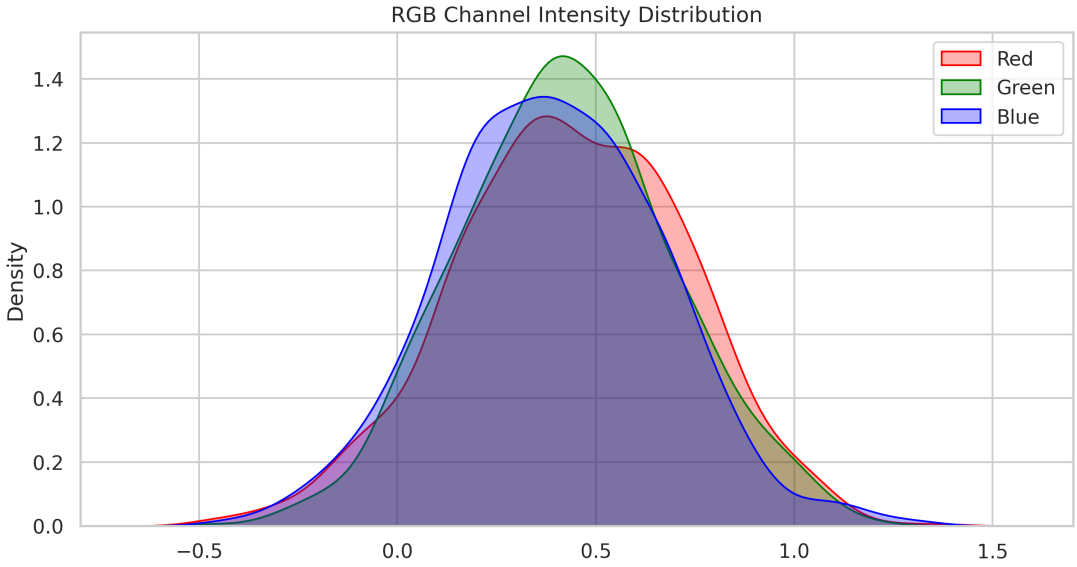



Figure 4.2: **Figure MM-2** — RGB Channel Pixel Intensity Distributions for a 500-image sample from Flickr8k. *Code:* `np.array(img).reshape(-1, 3)` for each image, then `plt.hist(channel_values, bins=50)` per channel.

- **Broader, more multi-modal distributions:** Unlike CIFAR-10's relatively flat distributions, Flickr8k shows peaks at both the low end (dark scenes, shadows) and high end (bright skies, reflective surfaces), reflecting the much greater photometric diversity of real-world photography.
- **Warm bias persists:** The Red channel tends to have a slightly higher mean than Blue, consistent with natural outdoor photography.
- **Normalization reference:** ImageNet mean/std normalization (`mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]`) is appropriate for this dataset when using pre-trained models.

## 4.3 Text Modality Analysis

### Caption Length Distribution (Character Level)

Figure MM-3 below shows a box plot of caption lengths in characters (all captions pooled). This provides a first look at the distribution of caption verbosity.

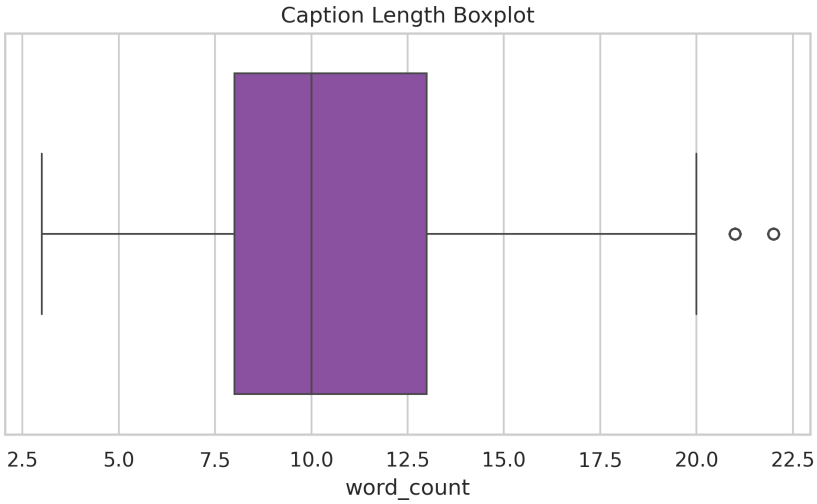



Figure 4.3: **Figure MM-3** — Caption Character Length — Box Plot for all 40,460 captions in Flickr8k. *Code:* `df['cap_len'] = df['caption'].apply(len)`, then `plt.boxplot()`.

- The interquartile range is compact, indicating consistent caption lengths across annotators.
- Few extreme outliers exist at both ends — annotators consistently produced moderate-length descriptions.

### Caption Word Count Distribution

Figure MM-4 below breaks caption length down to word (token) count. This is the operationally relevant metric for sequence modelling.

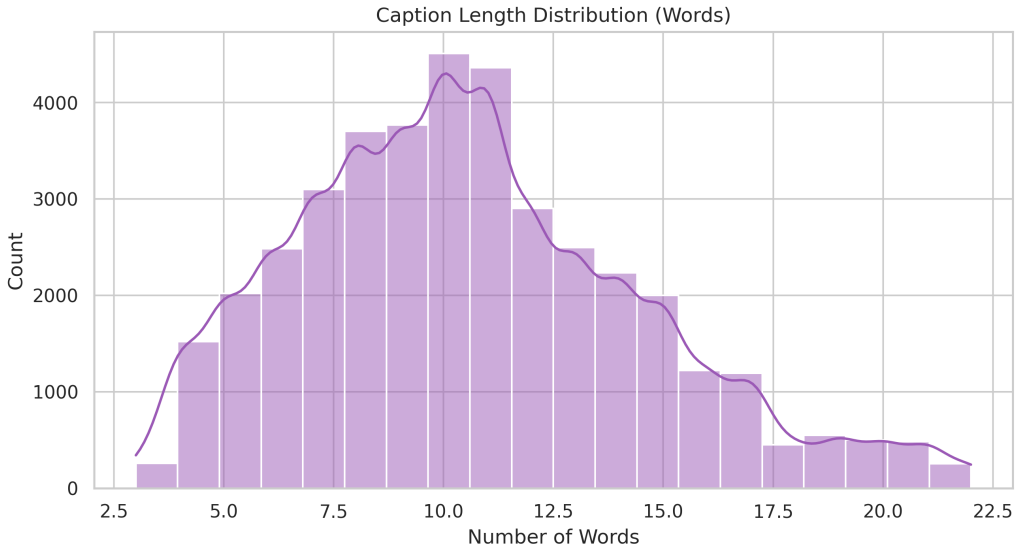



Figure 4.4: **Figure MM-4** — Caption Word Count Distribution across 40,460 captions. The distribution is unimodal and approximately normal, peaking at 10–11 words. *Code:* `df['n_words'] = df['caption'].str.split().str.len()`, then `plt.hist()`.

Key observations:

- Caption word counts follow a roughly **normal distribution** centred at ∼10–11 words per caption.
- The vast majority (90%) of captions fall in the range **8–15 words**; virtually none exceed 25 words.
- **Practical implication:** A `max_length = 20` tokens covers ∼90% of captions; `max_length = 25` covers ∼97%, making padding overhead minimal.
- The tight distribution means no caption domination by extreme outliers — the training signal is well-balanced across caption lengths.

### Top Caption Vocabulary

Figure MM-5 below shows the 20 most frequent content words after stopword removal. The vocabulary is **human-centric and action-rich**:

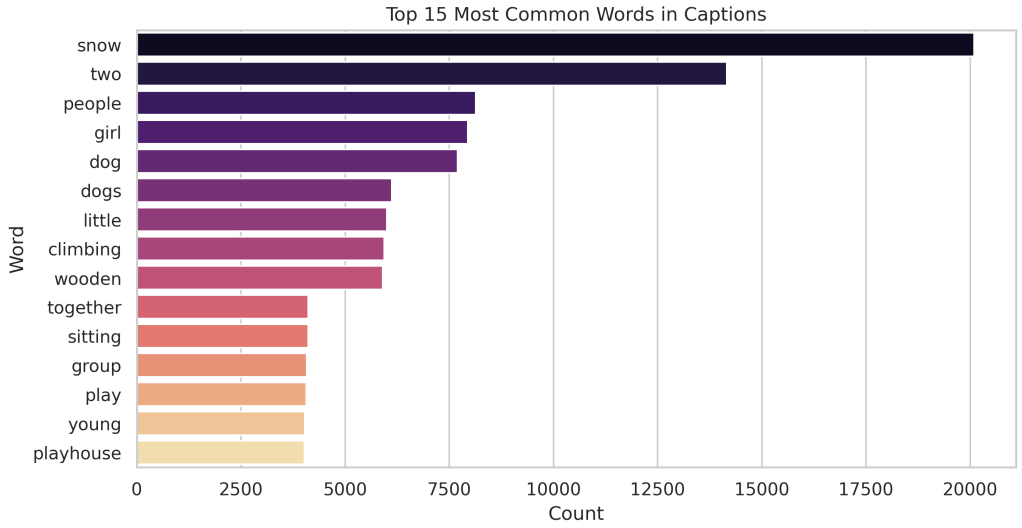



Figure 4.5: **Figure MM-5** — Top 20 Caption Words after Stopword Removal. Generated by `Counter` on all tokenised captions with a standard English stopword list applied.

- **Subject nouns dominate:** *dog*, *man*, *woman*, *boy*, *child*, *people* — Flickr8k is rich in photographs of people and animals.
- **Action verbs are frequent:** *running*, *playing*, *standing*, *walking* — the captions describe dynamic, real-world activities.
- **Colour and material descriptors:** *white*, *black*, *red*, *water* — spatial and visual context is consistently described.
- **No abstract vocabulary:** The text is grounded exclusively in observable, tangible scenes — ideal for cross-modal alignment.

### Word Count vs. Character Count in Captions

Figure MM-6 below plots word count against character count for all captions, confirming the very tight, well-controlled nature of the annotation process:

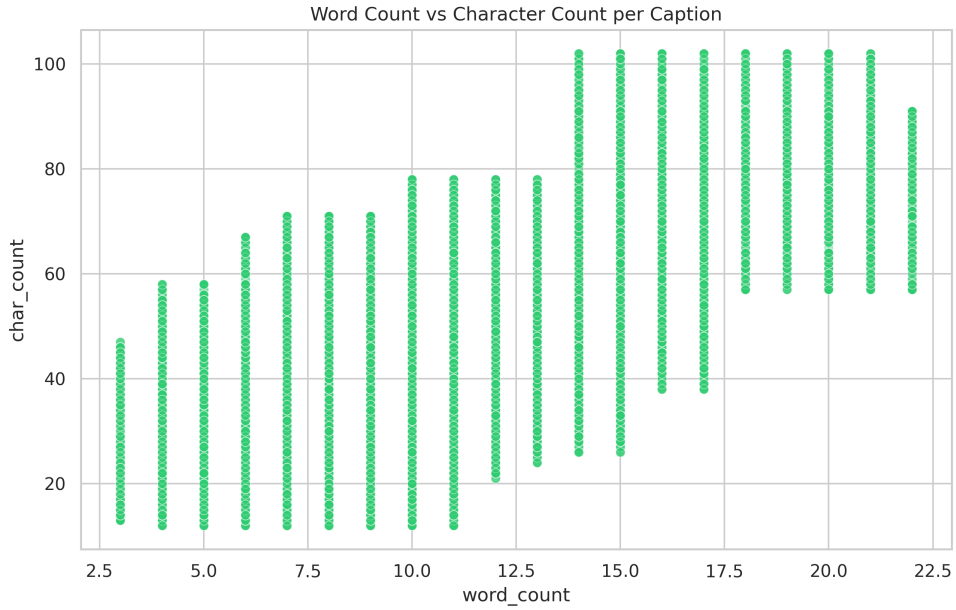



Figure 4.6: **Figure MM-6** — Word Count vs. Character Count for Flickr8k Captions. The near-linear relationship confirms consistent writing style across all 5 annotators per image. *Code:* `plt.scatter(df['n_words'], df['cap_len']).`

The tight, near-linear scatter (no outlier clouds) confirms that all 5 annotators per image maintained a **consistent linguistic register**: short words (∼4–6 chars each on average), standard sentence structure, and no extreme verbosity. This homogeneity is beneficial for model training — the caption space is predictable and bounded.

## 4.4 Key Findings and Preprocessing Recommendations

**Summary of Multimodal EDA Key Findings:**

- **Dual-modality structure:** 8,092 images × 5 captions each = 40,460 caption-image pairs. Perfect 1:5 alignment, zero missing data. High data integrity confirmed.
- **Non-uniform image resolution:** Images range widely in size (most longest-edge = 500 px). Uniform spatial preprocessing is mandatory before any CNN or ViT backbone.
- **Richer pixel diversity than CIFAR-10:** RGB distributions are broader and multi-modal, reflecting real-world photographic diversity. Use ImageNet normalisation constants when using pre-trained encoders.
- **Short, consistent captions:** 90% of captions are 8–15 words. `max_length = 25` tokens provides ∼97% coverage with minimal padding waste.
- **Human-centric vocabulary:** Dominant content words (*dog, man, woman, boy, running, playing*) reflect the dataset's bias toward people and animals in outdoor, dynamic settings.
- **Recommended preprocessing pipeline:**
  1. **Images**: Resize with aspect-ratio-preserving padding → $224 \times 224$ crop → normalize with ImageNet mean/std.
  2. **Captions**: Lowercase → tokenise (BPE or word-level) → add `[BOS]` / `[EOS]` tokens → pad/truncate at `max_length=25`.
  3. **Alignment**: Build an index mapping each image to its 5 captions; during training, randomly sample 1 caption per image per epoch for data augmentation.
- **Recommended models: CLIP** (OpenAI) for zero-shot cross-modal retrieval; **BLIP** or **GIT** for fine-tuned image captioning on this dataset.

# Project Links & Resources

This report is accompanied by a fully interactive web-based EDA Report, open-source code, and a presentation video. Please visit the following links for full details:

- **Interactive Landing Page:** https://dangkhoi-dev.github.io/P4AI-DS/
- **Source Code Repository:** https://github.com/dangkhoi-dev/P4AI-DS
- **Presentation Video (YouTube):** https://youtu.be/Ikld5pqy6e8

# Attribution & Data Sources

In compliance with data usage policies and academic integrity, we would like to acknowledge the original authors, creators, and curators of the datasets used in this Exploratory Data Analysis report. The datasets were sourced from the following repositories:

- **Titanic Dataset (Tabular Data):**
  Sourced from Kaggle. This dataset provides passenger information and survival status from the Titanic disaster.
  **Link:** https://www.kaggle.com/datasets/yasserh/titanic-dataset
- **SMS Spam Collection (Text Data):**
  Sourced from Kaggle. A public set of SMS labeled messages that have been collected for mobile phone spam research.
  **Link:** https://www.kaggle.com/datasets/marslinoedward/sms-spam-dataset/data
- **CIFAR-10 Dataset (Image Data):**
  Collected by Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton. The dataset consists of 60,000 32x32 color images in 10 classes.
  **Link:** https://www.cs.toronto.edu/~kriz/cifar.html
- **Flickr8k Dataset (Multimodal Data):**
  Sourced from Kaggle. A standard benchmark dataset containing 8,000 images, each paired with 5 different human-annotated captions.
  **Link:** https://www.kaggle.com/datasets/adityajn105/flickr8k